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RESEARCH ARTICLE

Federated Learning With AutoAlbum for Kidney Disease Detection: A Privacy-Preserving Approach to Medical Image Analysis

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ABSTRACT The increasing prevalence of kidney-related diseases necessitates efficient and accurate automated diagnostic tools to overcome the limitations of time-consuming and subjective manual image analysis. This study introduces a federated learning framework for decentralized kidney disease detection from CT scans, using AutoAlbum, an automated data augmentation technique, to introduce realistic variations through a diverse set of 11 augmentations. AutoAlbum enhances data variability and simulates real-world conditions, improving model robustness. The original Kidney Dataset (KD) is augmented globally using AutoAlbum to create a balanced dataset (AKD) for pre-trained model selection. KD is also partitioned into distributed subsets (KD-D) to mimic decentralized data in federated learning. Furthermore, augmentations are applied to these distributed KD subsets, resulting in the augmented and distributed dataset (AD-D), designed to evaluate the federated model's performance under Non-Identical and Independent Distribution (Non-IID) conditions. KD represents the original data, AKD the globally augmented data, KD-D the distributed original data, and AD-D the distributed and augmented data. The federated learning framework enables privacy-preserving collaborative training across KD-D and AD-D. Evaluations showed steady performance improvements on KD-D, reaching 0.9663 accuracy by round five. Notably, the model demonstrated remarkable robustness on the more challenging AD-D, achieving an exceptional accuracy of 0.9916 by the fifth round, highlighting the framework's potential for handling data heterogeneity in real-world clinical settings. To support model interpretability and clinical relevance, Grad-CAM visualizations were employed to reveal class-specific activation regions, enhancing transparency in model decision-making.

INDEX TERMS Deep learning, federated learning, image augmentation, kidney disease, pre-trained model, Grad-CAM.

I. INTRODUCTION

Human kidney disease impairs the kidneys' vital filtering functions, potentially leading to life-threatening complications. Common conditions include cysts, tumors, and stones. Simple kidney cysts are often asymptomatic, while Polycystic Kidney Disease (PKD) affects around 600,000 in the US and is a leading cause of kidney failure, with 25% experiencing initial cardiac symptoms [1]. Kidney tumors rank among the top 10 US cancers, with an estimated 81,610 new cases and 14,390 deaths projected in 2024, predominantly affecting older men and showing higher prevalence in certain ethnic groups [2]. Kidney stones affect about one in ten individuals, with men slightly more susceptible; these result from crystallized urinary chemicals and are linked to conditions like high blood pressure, diabetes, and obesity, causing severe symptoms [3]. While simple cysts are often benign, PKD and large cysts can lead to kidney failure, pain, obstruction, and rupture [4]. Kidney tumors, especially cancerous ones, impair kidney function and can metastasize [5]. Kidney stones can cause infections, hydronephrosis, and sepsis [6]. Early detection is crucial for managing all these conditions.

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Computed Tomography (CT) scans are often the preferred imaging technique for assessing kidney health due to their exceptional cross-sectional images providing a comprehensive view of kidney anatomy, including internal structures, surrounding tissues, and blood vessels [7]. This detailed visualization is crucial for accurately identifying abnormalities like tumors, cysts, stones, and obstructions, and for detecting subtle changes in kidney size, shape, and density indicative of early disease or progression. The detailed information from CT scans is invaluable for guiding treatment planning, such as determining kidney stone size and location for appropriate procedures or assisting in surgical planning for kidney tumors. Furthermore, CT scans effectively monitor treatment effectiveness for various kidney conditions. While ultrasound and Magnetic Resonance Imaging (MRI) have their advantages, CT scans frequently offer the most comprehensive and detailed information for kidney health evaluation, making them essential for diagnosis, treatment planning, and disease management.

Traditional kidney disease detection, relying on manual image interpretation and lab results, is prone to human error due to physician subjectivity and can be time-consuming, potentially delaying treatment [8]. These methods may also miss subtle disease indicators, highlighting the need for automated systems. Deep learning, particularly Convolutional Neural Networks (CNNs) [9], offers advantages over traditional machine learning by automatically learning complex features from raw images, enabling the detection of subtle patterns with higher accuracy for various kidney diseases. Furthermore, deep learning models can effectively process the complexity of medical images, improving diagnostic performance. Pre-trained deep learning models use knowledge from vast datasets, allowing for the learning of generalizable features [10]. Utilizing these models for kidney disease detection can significantly reduce the required training data, accelerate the training process, and potentially enhance performance by using already learned information.

Despite the benefits of pre-trained models like improved accuracy and faster training, their traditional use requires centralized data aggregation, raising privacy concerns for sensitive medical images. Federated learning addresses this by enabling collaborative training across institutions without sharing raw patient data [11]. Each institution trains a local model on its data, sharing only updated model parameters with a central server for global model aggregation. This decentralized approach keeps sensitive data private within each institution, enhancing security and potentially improving model generalizability by using diverse patient data.

Recognizing the limitations of traditional kidney disease detection from CT images, this study proposes a new approach addressing noise and data heterogeneity. To enhance model robustness against common CT scan noise, an augmented dataset incorporating 11 noise types was generated from high-quality images. A rigorous pre-trained model evaluation identified the optimal base for subsequent federated learning. Addressing data heterogeneity [11] (Non-Identical and Independent Distribution (Non-IID) data and data asymmetry) inherent across institutions, a federated learning framework was employed for collaborative training without centralizing sensitive patient data. This approach enables learning from diverse, non-uniformly distributed datasets while preserving privacy, crucial in real-world healthcare with varying disease prevalence and imaging protocols. The global model's performance was evaluated using precision, accuracy, F1-score, and recall, showcasing the framework's effectiveness in handling real-world data complexities for accurate kidney disease diagnosis.

In medical image analysis, particularly in critical domains like kidney disease detection, model transparency and interpretability are essential for clinical acceptance. Deep learning models often function as "black boxes," making it challenging for clinicians to understand the rationale behind the predictions. This limitation is even more pronounced in federated learning environments, where the decentralized nature of data restricts centralized validation and visibility into model behavior. Therefore, providing visual explanations that reveal how and why a model arrives at its decisions is crucial for building trust, ensuring clinical relevance, and enabling responsible AI deployment in healthcare. To address this need, we incorporated Grad-CAM visualizations in our study, highlighting the discriminative regions that the central model focused on when making predictions across different classes in the AD–D dataset.

The key contributions of this study are as follows:

A. MITIGATING NON-IID AND DATA ASYMMETRY CHALLENGES THROUGH DATA AUGMENTATION

This study incorporates AutoAlbum, an automated augmentation technique, to introduce realistic variations into kidney CT images, primarily to address the challenges posed by Non-IID data in federated learning. Non-IID data can degrade federated learning performance by causing local models to diverge, hindering global model convergence and generalization, and allowing institutions with larger datasets to disproportionately influence the global model. Data asymmetry can also slow convergence and introduce biases, favoring data from larger institutions and leading to performance disparities across patient populations. To mitigate these issues, AutoAlbum dynamically applies combinations of 11 diverse augmentation techniques—including Poisson Noise, Gaussian Noise, Speckle Noise, Salt-and-Pepper Noise, Flips, Brightness/Contrast, Gamma, and Blurs—to the original dataset (KD), generating an augmented dataset (AKD) with enhanced variability and realism. This approach overcomes limitations of traditional augmentation by introducing more diverse and realistic image transformations, ultimately enhancing model robustness and generalization.

B. CONSTRUCTION OF DISTRIBUTED DATASETS

The construction of distributed datasets, KD – D and AD – D, is crucial for simulating real-world federated learning scenarios. These datasets were created by partitioning the original

KD dataset into distinct subsets (clients), mirroring the decentralized nature of data distribution in federated learning. The KD – D dataset, consisting of non-diverse images, simulates the scenario where each subset represents the unique data available at a specific client. The AD – D dataset was created by applying the AutoAlbum augmentation strategy to each of the distributed subsets, introducing realistic variations and enhancing data diversity within each client's local dataset. These distributed datasets play a pivotal role in evaluating the robustness and generalizability of the federated learning model in handling Non-IID data distributions, a common challenge in real-world scenarios where data across different institutions exhibits significant variations in quality, quantity, and distribution.

C. PROPOSED ROBUST FEDERATED LEARNING FRAMEWORK FOR DECENTRALIZED KIDNEY DISEASE DETECTION

This study introduces a decentralized and collaborative federated learning framework, empowering clinicians to use private data while ensuring patient privacy. Despite inherent challenges like Non-IID distributions and data asymmetry across institutions, our proposed framework demonstrates robust performance in kidney disease detection, achieving results highly competitive with centralized approaches. Through rigorous experimentation and comparative analyses, we underscore the significance and practical utility of this strategy, particularly within critical medical domains.

D. VISUAL INTERPRETABILITY WITH GRAD-CAM

We provide Grad-CAM–based interpretability analysis by visualizing class-specific activation maps for correctly identified samples from the KD-D and AD–D dataset. These heatmaps highlight the regions most influential in the model's predictions and demonstrate that the model attends to clinically relevant features. This not only enhances transparency but also supports model validation in a federated learning context, where direct access to all client data is restricted.

This article proceeds as follows: A review of the current literature and existing approaches is presented in the ''Related Works'' section. The ''Materials & Methods'' section provides a detailed description of the proposed methodology. The ''Results'' section reports the findings and analysis obtained. A discussion of these results is presented in the ''Discussion'' section. Finally, the ''Conclusions'' section summarizes the key findings and outlines potential future research directions.

II. RELATED WORKS

The application of deep learning in kidney disease detection and classification has led to notable progress in recent years. CNNs, in particular, have shown strong performance, with studies leveraging techniques like data augmentation, transfer learning, and ensemble approaches to improve model reliability and boost classification accuracy.

The study by [12] develops an AI-based system for accurate kidney disease prediction and classification using a dataset of 12,446 images. Preprocessing involves noise removal and image enhancement, followed by segmentation using Otsu's binarization, distance transform, and watershed transformation. Contour features (area, intensity, etc.) are extracted from the segmented regions. Different deep learning models (DenseNet201, EfficientNetB0, InceptionResNetV2, MobileNetv2, ResNet50V2, and Xception) are trained with RMSprop, SGD, and Adam optimizers. The Xception model with RMSprop achieved the highest accuracy (99.89%), while ResNet50V2 (with SGD) and DenseNet201 (with Adam) also demonstrated best performance (99.68%). The study by [13] introduces ''IR-CNN,'' an efficient architecture based on InceptionResNetV2 for detecting kidney tumors, stones, and cysts in CT images. The model is customized with Global Average Pooling, Batch Normalization, dropout, and dense layers with swish activation for improved feature extraction and accuracy. Trained and tested on a 4000-image dataset, IR-CNN achieved 99.38% accuracy with Adam optimizer, surpassing 94.63% (SGD) and 97.38% (RMSprop). With only 5 misclassifications out of 800 test images, IR-CNN using Adam demonstrated the best performance. The study in [14] focused on classifying kidney conditions (cysts, normal, stone, tumor) using CT scan images. A deep learning framework incorporating spatial attention modules was developed to enhance classification accuracy and interpretability. By integrating these modules within a CNN architecture, the model gained the ability to identify and prioritize the most relevant regions within the CT images, providing valuable insights into the decision-making process. This attention mechanism significantly improved the model's performance, achieving an accuracy of 99.52%. Furthermore, the model demonstrated good precision (99.54%), recall (99.52%), and F1-score (99.52%), indicating its ability to identify and classify different kidney conditions. The research in [15] introduces a deep learning-based CNN model for the classification and prognosis of kidney diseases. Utilizing a benchmark CT kidney dataset, the study employs a multi-stage approach. Initially, the dataset undergoes pre-processing, encompassing grayscale conversion, image scaling, and bilinear interpolation to enhance image quality and uniformity. Subsequently, the CNN architecture extracts salient features from the pre-processed images. The final stage involves the classification of kidney conditions (tumor, cyst, normal, or stone) using the extracted features. The proposed approach demonstrates the performance of achieving a good accuracy of 0.992%, precision of 0.994%, recall of 0.982%, and F1-score of 0.987%, indicating its efficacy in classifying kidney diseases. The research in [16] introduces a novel computer-aided diagnosis model for kidney disease classification utilizing a modified Xception architecture. Using transfer learning with pre-trained ImageNet weights, the Xception model is fine-tuned on a dataset of 12,446 CT images (abdomen and urogram). This approach achieved the

performance, demonstrating an accuracy of 99.39%, precision of 99.33%, recall of 98%, and F1-score of 98.67% in lung cancer multi-class classification, highlighting its potential for kidney disease detection. The study by [17] investigates the classification of kidney CT images into four categories: cyst, normal, stone, and tumor, employing three distinct CNN-based approaches. In the initial stage, watershed segmentation is utilized to isolate the region of interest within the CT images. Subsequently, the segmented data is employed to train three classification networks: EAnet, a pre-trained ResNet50 model, and a custom-designed CNN architecture. Training and evaluation are conducted using the publicly available CT Kidney Normal Cyst Tumor and Stone dataset from Kaggle. The custom CNN model demonstrated the performance, achieving an accuracy of 98.66% on the test set, surpassing the accuracies of EAnet (83.65%) and ResNet50 (87.92%). The study by [18] introduces SpinalZFNet, a hybrid deep-learning architecture that integrates the strengths of SpinalNet and ZFNet for accurate kidney disease classification using CT images. This synergistic approach enhances feature analysis, leading to better classification performance while optimizing computational efficiency. The proposed framework incorporates a multi-step process. Initially, acquired CT images undergo pre-processing using a median filter, followed by segmentation using Efficient Neural Network (ENet). Subsequently, image augmentation techniques are applied to enhance data diversity. The extracted features from the augmented images are then utilized by SpinalZFNet to classify kidney conditions into normal, tumor, cyst, and stone categories. SpinalZFNet demonstrated good performance, achieving 99.9% sensitivity, 99.5% specificity, 99.6% precision, 99.8% accuracy, and 99.7% F1-Score. The research by [19] introduces two novel ensemble models, StackedEnsembleNet and PSOWeightedAvgNet, for improved kidney stone detection in CT images. StackedEnsembleNet integrates InceptionV3, InceptionResNetV2, MobileNet, and Xception through a two-level deep stack ensemble, while PSOWeightedAvgNet employs Particle Swarm Optimization to determine optimal weights for a weighted average ensemble. Evaluated on a dataset of 1799 CT images, both ensemble models surpass the individual base models, achieving high accuracy rates. Further validation on an unseen dataset and comparisons with existing methods confirm their superior performance with 98% accuracy. The study also includes Grad-CAM visualizations and error case analysis to provide insights into the models' decision-making processes. The study by [20] introduces SSLSD-KTD, a masked autoencoder framework for kidney tumor detection. Using self-supervised learning through self-distillation, SSLSD-KTD incorporates masked patches and calculates losses on both encoder and decoder outputs. By combining local attention in the encoder with global attention in the decoder, SSLSD-KTD achieves results, attaining 98.04% accuracy on the KAUH-kidney dataset (8400 samples) and 82.14% on the CT-kidney dataset (840 samples). Further enhancement through transfer learning by incorporating external information boosts accuracy to 99.82% and 95.24% on the respective datasets. The study by [21] proposes F2 HCN2, a feature fusion hypergraph CNN framework for early diagnosis of kidney stones, cysts, and tumors from CT images. This approach uses discriminative features extracted from DarkNet19 and residual features extracted from ResNet50. These features are then integrated and classified using hypergraph representation learning. The proposed model is trained on a comprehensive dataset of 12,446 CT images, and its performance is evaluated using metrics such as accuracy, F1-score, recall, PPV, and AUC obtaining an accuracy of 99.71%. The research in [22] proposes a deep end-to-end CNN model to enhance the detection and characterization of kidney problems in CT scans. Trained on a standard CT image library annotated for kidney stones, cysts, and tumors, the model achieves a detection rate of 99.17% in validation tests. Further evaluation on a larger dataset with 14,146 samples demonstrates an accuracy of 99.68%. Finally, validation on a clinical dataset yields 99% accuracy. The study by [23] introduces "StoneNet," a lightweight model for kidney stone detection in CT images, based on MobileNet with depthwise separable convolutions. Incorporating Global Average Pooling (GAP) instead of flattening layers significantly enhances model robustness and reduces the number of parameters. The model's performance demonstrates an accuracy of 97.98% with significantly reduced training and testing times of 996.88 s and 14.62 s, respectively. Extensive experimentation with different batch sizes and optimizers validates StoneNet's performance and computational efficiency. Evaluated on a large kidney dataset of 1799 CT images. The paper by [24] introduces a hybrid ensemble framework for improved kidney tumor classification and segmentation. The framework integrates a U-Net architecture incorporating visual capsule networks with a deep Feed-Forward Extreme Learning Machine (FFELM) classifier. To enhance performance, the framework incorporates data augmentation, Saliency Tumor Segmentation (STS), and a multi-level classification scheme based on FFELM. Extensive experimentation demonstrates the performance of this hybrid approach, achieving a Dice coefficient of 0.96 and an accuracy of 97.5% for kidney tumor classification. The research by [25] introduces an approach for accurate and quantitative diagnosis of chronic kidney cyst diseases through CT image segmentation. The study utilizes a deep learning framework incorporating the Residual Dual-Attention (RDA) module to segment renal cysts in CT scans of 79 patients, with 52 cases for training and 27 for testing. The RDA-UNET model demonstrated rapid convergence and achieved high segmentation accuracy, with Dice Similarity Coefficients (DSC) of 96.25% and 94.22% for the left and right kidneys, respectively. Furthermore, the model achieved 96.34% precision, and 96.88% recall for the left kidney and 95.34% precision, and 94.61% recall for the right kidney cyst detection. The study by [26] introduces "KidneyMultiNet," a

framework that combines Densenet201 and Xception convolutional neural networks within a transfer learning paradigm for accurate kidney disease prediction based on CT scan images. The proposed framework demonstrates the potential for reliable and thorough kidney disease diagnosis. Utilizing a publicly available kidney CT scan image dataset, the model was trained using a train generator with a learning rate schedule to optimize performance. After extensive training with numerous kidney images, KidneyMultiNet achieved a maximum accuracy of 99.92%.

A. LIMITATIONS AND CHALLENGES IN EXISTING MODELS

The major limitations noticed in the previous studies are as follows. Previous studies on data augmentation for medical imaging, particularly in the context of kidney disease detection from CT scans, have exhibited several limitations. Firstly, traditional augmentation techniques often fall short of capturing the full spectrum of real-world variations that can occur in CT images. CT scans are susceptible to various types of noise, such as Gaussian noise, Poisson noise, and speckle noise, which can significantly impact image quality and diagnostic accuracy. Traditional augmentation methods may not adequately simulate these noise patterns, limiting the model's ability to generalize to real-world clinical scenarios. Artifacts, such as motion artifacts, metal artifacts, and beam hardening artifacts, can frequently occur during CT image acquisition. These artifacts can significantly degrade image quality and introduce misleading information. Traditional augmentation techniques cannot often effectively simulate these artifacts, hindering the model's robustness in real-world clinical settings. Variations in patient positioning, breathing patterns, and body habitus can introduce significant variability in CT images. Traditional augmentation techniques may not adequately capture these patient-specific variations, potentially limiting the model's ability to generalize across different patient populations. The appearance of kidney diseases can vary significantly depending on the specific pathology, disease stage, and patient characteristics. Traditional augmentation techniques may not effectively simulate these intra-class variations, potentially hindering the model's ability to accurately classify and diagnose different types of kidney diseases.

The previous approaches however may be efficient but often necessitate the centralization of patient data from multiple institutions for model training. This centralization poses significant privacy risks, as it concentrates sensitive patient information in a single location, making it vulnerable to potential breaches, cyberattacks, and unauthorized access. A single data breach at the central repository can compromise the privacy of a large number of patients, potentially leading to severe legal and reputational consequences for the involved institutions. The reluctance of healthcare institutions to share sensitive patient data can create significant data silos, hindering the development of robust and generalizable AI models. The lack of data sharing can limit the diversity and representativeness of training datasets, leading to biased models that may not generalize well to diverse patient populations. Limited data availability can significantly hinder research progress in medical imaging, as researchers may not have access to the large and diverse datasets required to develop cutting-edge AI models.

B. SOLUTION USING THE DEVELOPED MODEL

This study addresses the limitations of existing approaches by incorporating two new strategies. Firstly, the integration of AutoAlbum, an automated augmentation technique, within the federated learning framework effectively mitigates the challenges of Non-IID data and data asymmetry. By introducing diverse and realistic augmentations such as Poisson Noise, Gaussian Noise, Speckle Noise, Salt-and-Pepper Noise, various Flips, Random Brightness & Contrast, Random Gamma, Gaussian Blur, and Motion Blur, AutoAlbum improves the model's ability to handle variations in image characteristics, thereby enhancing its generalization to diverse data distributions typically found in real-world clinical environments. Secondly, the federated learning framework itself provides a robust mechanism for handling data heterogeneity and variations across different institutions. By enabling collaborative model training while preserving data privacy, federated learning allows the model to learn from diverse data distributions, improving its generalization capabilities and mitigating the impact of data variations observed in real-world clinical scenarios.

III. PROPOSED APPROACH

This section delves into the intricacies of our proposed framework for kidney disease detection using deep learning within a federated learning paradigm. We commence with the dataset description (both original and augmented), a foundational overview of the federated learning framework, followed by a detailed description of our proposed architecture. Subsequently, we elucidate the client-side model training procedure and the server-side model aggregation process, providing a comprehensive understanding of the proposed methodology. Figure 1 represents the block diagram of the proposed approach.

A. DATASET

1) ORIGINAL (PUBLIC) DATASET

This study utilized a comprehensive dataset of 12,446 unique CT images acquired from the Picture Archiving and Communication System (PACS) of various hospitals in Dhaka, Bangladesh [27]. The dataset encompasses patients with confirmed diagnoses of kidney cysts, tumors, stones, and normal findings. Both coronal and axial cuts from both contrast and non-contrast studies of the whole abdomen and urogram were included. A rigorous selection process involved careful manual review by radiologists and medical technologists to ensure the accuracy of each diagnosis. Following data anonymization and conversion to lossless jpg format, the final dataset comprised 3,709 cysts, 5,077 normal, 1,377 stone, and

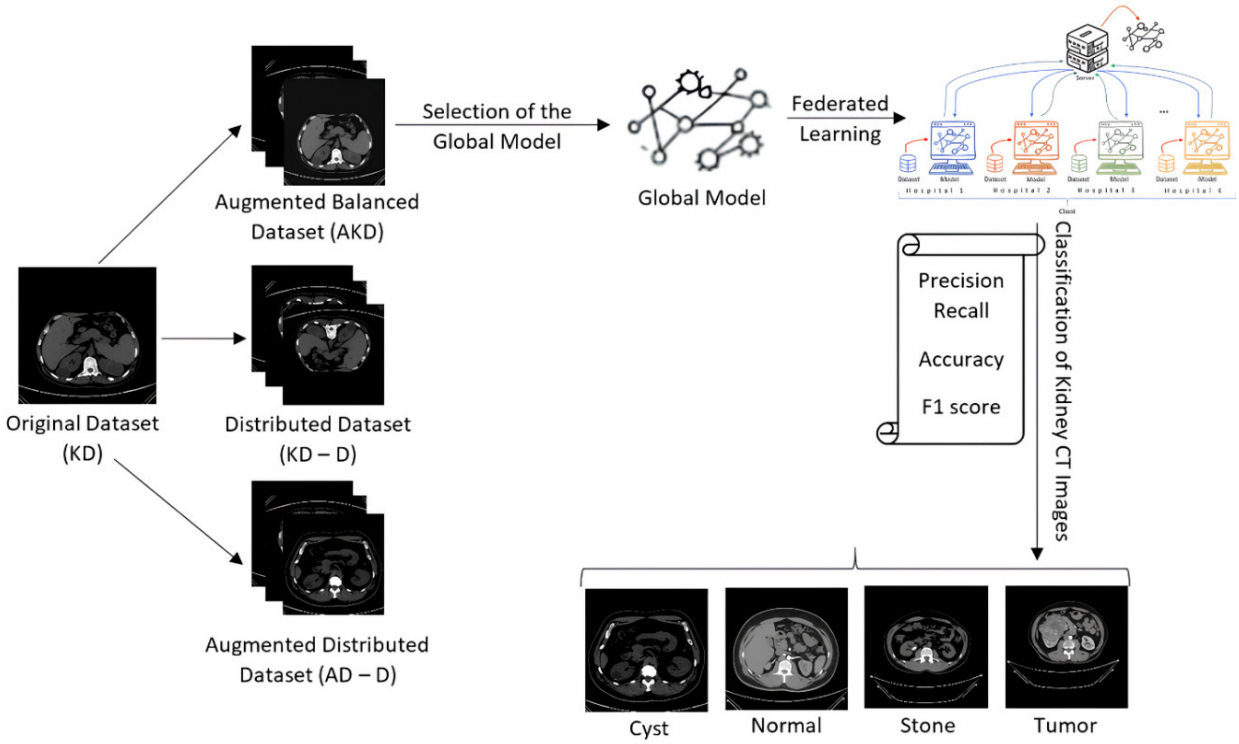


FIGURE 1. The block diagram of the proposed approach.

2,283 tumor images, providing a robust foundation for the subsequent analysis. In this study, this dataset is named as KD dataset.

2) AUGMENTED DATASET

This study incorporates AutoAlbum, an automated augmentation technique, to introduce realistic variations into the kidney CT images. Recognizing the limitations of traditional augmentation methods in capturing the full spectrum of real-world variations in CT scans, such as noise, artifacts, and patient-specific variability, AutoAlbum employs a diverse repertoire of 11 augmentation techniques, including Poisson Noise, Gaussian Noise, Speckle Noise, Salt-and-Pepper Noise, Vertical Flip, Horizontal Flip, Horizontal and Vertical Flip, Random Brightness & Contrast, Random Gamma, Gaussian Blur, and Motion Blur.

To further enhance the diversity and realism of the augmented dataset, AutoAlbum employs a probabilistic approach to dynamically select and apply unique combinations of these techniques to each image within the original dataset (KD). Specifically, a probability-based mechanism is implemented where, for each image, the number of augmentations to apply is randomly determined. For instance, the system might randomly select to apply 1, 2, 3, or more of the 11 augmentation techniques to a given image. This probabilistic approach introduces an element of randomness and variability in the augmentation process, resulting in a more diverse and representative augmented dataset (AKD) with 5000 images per class (cyst, tumor, stone, and normal). This balanced augmentation is crucial for mitigating potential biases and improving model generalization, particularly in the context of federated learning where data distribution across participating institutions can be highly heterogeneous.

By introducing more diverse and realistic image transformations, AutoAlbum effectively enhances the model's robustness and generalizability, enabling it to better handle the variations encountered in a real-world clinical environment, where CT images may demonstrate substantial differences in terms of noise, artifacts, and patient characteristics. This AKD dataset is crucial for determining the most suitable base pre-trained model that will serve as the foundation for our federated learning framework for kidney disease detection. The PseudoCode of the same is reported in Algorithm 1.

3) PUBLIC DISTRIBUTED DATASET

The public distributed dataset (KD – D) is crucial for establishing a robust federated learning framework. To simulate a real-world federated learning scenario, the original Kidney Dataset (KD) is divided into 'n' distinct subsets. By dividing the original dataset into subsets, we effectively simulate the heterogeneous data distribution observed in real-world federated learning settings. Each "client" will possess a unique subset of the data, reflecting the variations in disease prevalence, and imaging protocols that are commonly encountered in clinical practice. This approach adheres to the core principles of federated learning by keeping the data localized within each "client." This prevents the central

Algorithm 1 AutoAlbum Augmentation

```
Input:
  KD (original dataset of images)
  MaxAugmentations (maximum number of
augmentations per image)
  AugmentationPool = {Poisson Noise, Gaussian
Noise, Speckle Noise, Salt-and-Pepper Noise,
Vertical Flip, Horizontal Flip, Horizontal and
Vertical Flip, Random Brightness & Contrast,
Random Gamma, Gaussian Blur, Motion Blur}
Output:
  AKD (augmented dataset)
Initialize:
  - Define AugmentationPool containing
11 techniques.
  - Define MaxAugmentations (maximum number of
augmentations to apply per image).
  - Define uniform probability distribution for
the number of augmentations to apply.
Step 1: For each image in the original dataset KD:
1.1 Determine the Number of Augmentations:
  - Randomly select the number of augmentations
(numAugments) to apply to the current image, based
on a uniform probability distribution over the
range [0, MaxAugmentations].
  1.2 Select Augmentation Techniques:
      - Randomly select a subset of augmentation
techniques (subsetAugments) from the
AugmentationPool.
      - The size of the subset is equal to
numAugments, and the techniques are selected
without replacement.
  1.3 Apply Augmentations:
      - Sequentially apply each selected
augmentation technique in subsetAugments to the
original image.
Step 2: Output the augmented dataset AKD:
      - Generate the augmented dataset AKD
containing the original images and the augmented
images after applying the selected augmentations.
```

Algorithm 2 Creation of Public Distributed Dataset

```
Input:
  KD (original kidney disease dataset)
  n (number of clients)
Output:
  Public distributed dataset (KD - D)
Initialize:
  - Define KD (original kidney disease dataset).
  - Define n (number of clients).
  - Split KD into n distinct subsets (KD-D), each
representing the data for a client.
  - Ensure the subsets are heterogeneous,
reflecting variations in disease prevalence and
imaging protocols.
Step 1: Divide Original Dataset (KD) into n
Subsets:
   1.1. Divide KD into n subsets: {D1, D2, ...,
Dn}, ensuring heterogeneity:
      - Each subset Di contains a portion of the
kidney disease data, simulating different disease
prevalence and imaging protocols.
      - Each subset Di is localized for a specific
"client."
Step 2: Output:
       - THE KD - D DATASET
```

aggregation of sensitive patient data, thereby mitigating privacy risks and ensuring compliance with relevant data protection regulations. By creating multiple "clients," we enable collaborative model training across these simulated institutions. This allows us to evaluate the effectiveness of the federated learning framework in aggregating knowledge from diverse data sources while maintaining data privacy. The use of this KD – D dataset allows us to assess the robustness of the federated learning model in handling data asymmetry and Non-IID data distributions, which are common challenges in real-world federated learning scenarios. Notably, the KD – D dataset comprises high-quality images, further enhancing the rigor of the evaluation by minimizing the impact of image quality variations on model performance. The pseudo-code of the same is reported in Algorithm 2.

4) AUGMENTED DISTRIBUTED DATASET

The Augmented Distributed Dataset (AD – D) is constructed from the initial public dataset (KD). Notably, unlike the AKD dataset, which is a centralized and balanced augmented dataset, the AD – D dataset maintains the distributed nature of the data, mirroring the decentralized nature of federated learning. This distributed augmentation strategy is crucial for simulating the challenges of data heterogeneity and Non-IID data distributions often encountered in real-world federated learning scenarios. The 11 augmentation techniques (Poisson Noise, Gaussian Noise, Speckle Noise, Salt-and-Pepper Noise, Vertical Flip, Horizontal Flip, Horizontal and Vertical Flip, Random Brightness & Contrast, Random Gamma, Gaussian Blur, Motion Blur) are applied to KD dataset images to introduce realistic variations into the images, such as noise, artifacts, and other real-world distortions. First, the images in $Split_n$ are augmented using the AutoAlbum augmentation technique (as discussed in the Section "Augmented Dataset"). Then the augmented images from $Split_n$ are added to the original images within $Split_p$ of KD dataset. After that the augmented images from $Split_p$ of KD dataset are added to the original images within $Split_q$ of KD dataset and so on. This cascading approach ensures that each split receives augmented images from the preceding split, creating a more diverse and interconnected dataset across the clients. The AD – D dataset introduces controlled variations across the different splits, simulating the heterogeneous data distributions commonly encountered in real-world federated learning scenarios. By exposing the model to a wider range of data variations, the AD – D dataset enhances the model's robustness and generalizability, enabling it to better adapt to the diverse and challenging data distributions encountered in real-world clinical settings. The AD – D dataset ultimately contains approximately twice the number of images as the KD dataset, resulting in a total of 24,892 images. The pseudo-code of the same is reported in Algorithm 3.

Algorithm 3 Augmented Distributed Dataset (AD-D) Construction

```
Input:
  KD (original public dataset)
  AugmentationTechniques = {Poisson Noise,
Gaussian Noise, Speckle Noise, Salt-and-Pepper
Noise, Vertical Flip, Horizontal Flip, Horizontal
and Vertical Flip, Random Brightness & Contrast,
Random Gamma, Gaussian Blur, Motion Blur}
  n (number of splits or clients)
Output:
  AD-D (Augmented Distributed Dataset)
Initialize:
  - Define KD (original public dataset).
  - Define Augmentation Techniques (set of
11 augmentation techniques).
  - Define n (number of splits or clients for data
distribution).
Step 1: Augment Images in Split n:
   1.1. For each image in Split n of KD:
      - Randomly select a subset of
augmentation techniques (subsetAugments) from
AugmentationTechniques.
      - Apply the selected augmentations to the
image sequentially.
      - Store the augmented images.
Step 2: Add Augmented Images to Next Split:
   2.1. For each subsequent split (starting from
Split p to Split q):
      - Add the augmented images from Split n to
the original images in Split p of the KD dataset.
      - Then, add the augmented images from Split p
to the original images in Split q, and so on for
all splits.
Step 3: Construct Augmented Distributed Dataset
(AD-D):
   3.1. Collect all the augmented images from each
split and combine them with the original dataset.
   3.2. The final dataset (AD-D) now contains
both the original images and the augmented images
across all splits.
Step 4: Output:
      - Return the AD-D (Augmented Distributed
Dataset) containing both original and augmented
images from each split.
```

B. PROPOSED MODEL FOR KIDNEY DISEASE DETECTION

This study investigates a federated learning framework for kidney disease classification based on a client-server architecture (illustrated in Figure 2), employing the FedAvg algorithm. FedAvg or FederatedAveraging, which aims to minimize a generic non-convex neural network loss function, assumes a synchronous update scheme and is practically equivalent to a centralized model when trained on IID data [28]. Within this framework, a central server maintains a global model, which is shared with and updated by participating clients. By collaboratively training on their private datasets, clients contribute to the development of a powerful and robust model for kidney disease classification while maintaining data privacy and security.

This study introduces a Pre-Trained Deep Convolutional Neural Network (PT-DCNN) architecture specifically designed for the accurate classification of kidney diseases based on CT images. The proposed custom PT-DCNN model takes as input a CT image and outputs the probability of different kidney diseases, including cysts, tumors, and stones, as well as healthy kidney CT images. The intricacies of this custom PT-DCNN architecture, including its specific layers, connectivity patterns, and parameter configurations, are comprehensively detailed in the Section "Proposed Custom PT-DCNN."

This study employs a federated learning approach for kidney disease classification, where the learning process involves multiple communication rounds between a central server and participating clients. Before training, the custom PT-DCNN model is initialized with random weights. Assuming n clients, each possessing a dataset of D_n kidney images, each communication round (r) encompasses four distinct steps [29]:

Step 1: The federated learning process commences with the initialization of a global model (G) by the central server. This initial model, characterized by its initial weights (w^{r-1}), is subsequently disseminated to a selected subset of participating clients (in this study clients means different hospitals or clinics).

Step 2: In the second step, each participating hospital (client n) receives the initial model parameters (w^{r-1}) from the central server. Subsequently, each client conducts local training on a mini-batch of its private kidney image dataset. This local training process involves minimizing the categorical cross-entropy loss using the Adam optimizer with a locally defined learning rate for an optimized number of epochs obtained by using the EarlyStopping method. The main goal of the custom PT-DCNN model is to enhance the model's ability to accurately classify kidney diseases based on the local data.

Step 3: Upon completion of the local training phase, where each hospital (client n) has iteratively updated the model parameters on its local kidney image dataset, the updated model parameters (w_n^r) are transmitted back to the central server. This crucial step facilitates the accumulation of knowledge and improvements garnered from the local training processes across all participating clients or hospitals.

Step4: In the final step of the communication round, the central server receives the updated model parameters (w_n^r) from all participating hospitals. Subsequently, the server averages these individual updates to compute the new global model parameters (w^r) using a FedAvg scheme as defined in Eq. (1). This average process effectively combines the knowledge and improvements learned by each hospital on their local kidney image datasets, resulting in a refined global model that reflects the collective insights of the entire network of participating institutions.

$$w^r = \sum_{i=i}^{n} \frac{D_n}{n} \times w_n^r \tag{1}$$

This FedAvg considers the number of data points (D_n) contributed by each hospital (n), ensuring that hospitals with

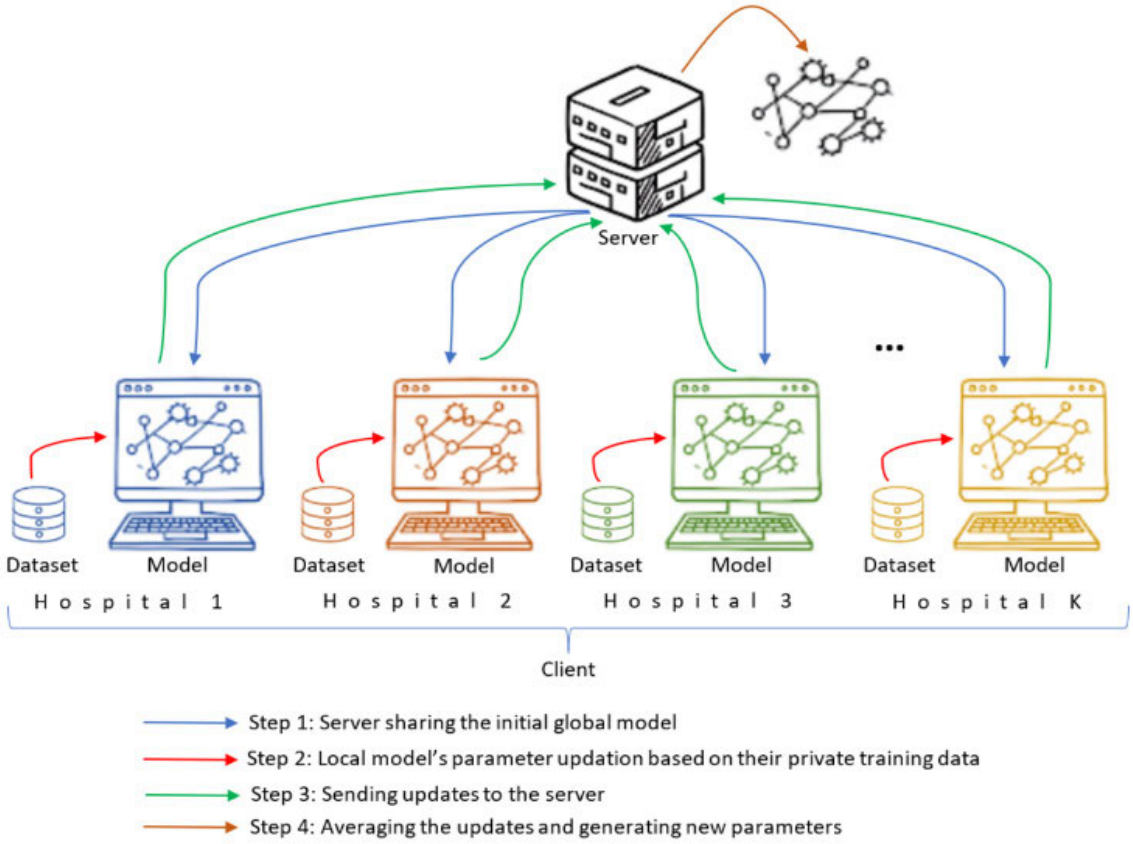


FIGURE 2. The federated learning framework for kidney disease classification used in this study.

larger datasets have a greater influence on the global model update. The new global model parameters are computed (w^r) using the FedAvg scheme.

The aforementioned four steps collectively constitute a single communication round within the federated learning framework for kidney disease classification. This iterative process is repeated multiple times, with the central server disseminating the updated global model parameters (w^{r-1}) obtained in the previous round (r-1) to a new subset of participating clients in each subsequent round. This dynamic client selection, detailed in the Section "Client-Side Training Process in the Federated Learning Framework," enhances the diversity of data utilized in the training process and further improves the robustness and generalization capabilities of the federated learning model.

1) PROPOSED CUSTOM PT-DCNN

This study introduces a collaborative and decentralized framework for the detection and classification of kidney diseases utilizing federated learning on a PT-DCNN. Recognizing the critical importance of data privacy, our primary objective is to demonstrate the efficacy of federated learning in using the rich information present in distributed datasets while safeguarding patient confidentiality. While the specific PT-DCNN architecture plays a secondary role in this investigation, we rigorously evaluated 21 pre-trained models on the augmented kidney dataset (AKD) to select the most suitable backbone network.

The selection of base models for federated learning was driven by the need for architectures that offer a balance of accuracy, computational efficiency, and suitability for distributed training. Models like VGG19, InceptionV1, InceptionV3, ResNet (50 – 152), and DenseNet (121 – 201) have demonstrated high accuracy in various image classification tasks, providing a strong foundation for medical image analysis. EfficientNetB (0 – 7) and MobileNetV2/V3 (Small/Large) are designed with computational efficiency in mind, making them suitable for resource-constrained environments and facilitating faster training and inference, which is crucial in federated learning where communication costs can be significant. These architectures have been extensively studied and optimized, making them well-suited for deep learning applications and readily adaptable to the federated learning framework. This diverse selection of base models allowed for a comprehensive evaluation of their performance within the federated learning paradigm, enabling the identification of the most suitable architecture for robust and efficient kidney disease classification.

Ultimately, EfficientNetB0 was chosen for its performance and efficiency. Our custom EfficientNetB0 architecture

employs the EfficientNetB0 backbone, excluding the original fully connected layer. A new classification head, comprising multiple dense layers and a final fully connected layer with softmax activation, is subsequently added to the backbone. Before appending the classification head to the base EfficientNetB0 model, GAP was employed. This architectural choice was motivated by several key advantages over traditional flattening [30]. GAP extracts global spatial information by averaging feature maps across spatial dimensions. This reduces the dimensionality of the feature representation while preserving the most important features, effectively mitigating the risk of overfitting, especially when dealing with limited data or complex image variations. By capturing global spatial information, GAP encourages the model to learn more generalizable features that are less sensitive to minor variations in image positioning or scale. This can lead to improved performance on unseen data and better generalization capabilities. GAP significantly reduces the number of parameters compared to flattening, leading to a more compact and computationally efficient model. This can be particularly beneficial in resource-constrained environments or when deploying the model on edge devices with limited computational power. GAP provides a more interpretable representation of the image, as it focuses on the most prominent features across the entire spatial domain. This can aid in understanding the model's decision-making process and identifying critical image regions for classification. This architecture accepts 224 × 224 pixel kidney images as input and produces four probability values corresponding to the four disease classes: cyst, tumor, stone, and normal. The model is optimized using categorical cross-entropy loss to maximize classification accuracy.

2) CLIENT-SIDE TRAINING PROCESS IN THE FEDERATED LEARNING FRAMEWORK

This study employs a federated learning approach where training is performed locally on each client device. Each of the four clients (used in this study) possesses a unique dataset of kidney images and identical CNN architectures as described in Section "Proposed Custom PT-DCNN." The training process commences with each client receiving the global model parameters (w^r) from the central server. Subsequently, each client independently trains its local model using these parameters, performing multiple iterations of optimization to minimize the loss function on its local dataset. Importantly, this decentralized training approach ensures that client data remains private and is never shared with the central server or other clients, thereby safeguarding patient confidentiality [31]. The pseudo-code of the same is reported in Algorithm 4.

Algorithm 4 Client-Side Training

```
Input:
    w^r (global model parameters)
    D_n (local dataset for client n)
    η (learning rate)
    E (number of epochs)
    B (batch size)
Initialize:
    w_n^r ← w^r # Set local model parameters to global
model parameters
Training Process:
    for epoch = 1 to E do:
      Shuffle D_n # Shuffle the local dataset for
this epoch
      # Iterate over mini-batches of the dataset
      for each mini-batch (X_batch, y_batch) in Dk
do:
        # Compute the loss for the current
mini-batch
        L ← loss(X_batch, y_batch, w_n^r)
        # Compute the gradients of the loss w.r.t.
local model parameters
        ∇L ← ∂L/∂w_n^r
        # Update local model parameters using Adam
optimizer
        w_n^r ← w_n^r - η * Adam(∇L)
    Output:
        Updated local model parameters: w_n^r
```

3) ROLE OF THE CENTRAL SERVER IN THE FEDERATED LEARNING FRAMEWORK

The central server plays a pivotal role in orchestrating the federated learning process for kidney disease classification. It serves as the custodian of the global model, initializing it with the initial set of parameters. The server then distributes this global model to each participating client, facilitating the commencement of local training. After each communication round, the server synchronously receives updated model parameters from all participating clients. Subsequently, the server employs Eq. (1) to average these individual updates, effectively combining the knowledge and improvements gained from the local training processes across all institutions. This aggregated information is then used to refine the global model parameters, driving the overall learning process towards improved accuracy and generalization in kidney disease classification. The federated learning process involved "n" rounds of communication and model updates. This iterative approach allows for gradual refinement of the global model through collaborative learning among the participating clients. By conducting multiple rounds of training and averaging, the federated learning framework enables the global model to progressively learn from the diverse data distributed across different clients, leading to improved generalization and enhanced performance on unseen data [32]. The PseudoCode of the same is reported in Algorithm 5.

C. PERFORMANCE MEASURES

To rigorously assess the performance of the federated learning model for kidney disease detection, a comprehensive evaluation framework was employed. This framework encompassed a suite of key performance metrics which are as follows [33]:

Algorithm 5 Server-Side Model Update

```
Input:
    w^r (global model parameters)
    {w_n^r} (updated model parameters from clients)
    {D_n} (number of data points at each client k)
    n (total number of data points across all
clients)
Output:
    w^{r+1} (updated global model parameters)
Algorithm:
    1. Receive Client Updates:
        - Receive updated model parameters (w_n^r) from
each participating client.
    2. Average Model Updates:
        - For each client n, calculate the FedAvg
model update:    w^{r+1} ← (1/n) * Σ (D_n * w_n^r) for all
clients n
    3. Return:
        - Updated global model parameters: w^{r+1}
```

Accuracy: Represents the overall correctness of the model's predictions, calculated as the ratio of correctly classified instances (True Positives (TP) + True Negatives (TN)) to the total number of instances (True Positives (TP) + True Negatives (TN) + False Positives (FP) + False Negatives (FN)).

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \tag{2}$$

Precision: Measures the proportion of true positive predictions among all positive predictions made by the model.

$$Precision = \frac{TP}{TP + FP} \tag{3}$$

Recall (Sensitivity): Measures the proportion of actual positive cases that were correctly identified by the model.

$$Recall = \frac{TP}{TP + FN} \tag{4}$$

F1-Score: Represents the harmonic mean of precision and recall, providing a balanced measure of the model's performance.

$$F1Score = 2 \times \frac{Precision \times Recall}{Precision + Recall} \tag{5}$$

These metrics provide a comprehensive evaluation of the model's ability to accurately detect and classify different types of kidney diseases, including cysts, tumors, stones, and normal cases. By analyzing these metrics, we can gain valuable insights into the model's strengths and weaknesses, identify areas for improvement, and ultimately enhance the reliability and effectiveness of the federated learning framework for kidney disease diagnosis.

IV. RESULTS AND ANALYSIS

This research used a high-performance computing environment to facilitate efficient and timely model training. The training process was executed on a system equipped with an NVIDIA GeForce RTX 3050 Ti Laptop GPU with 4GB VRAM, providing substantial computational power for accelerated deep learning operations. This GPU, coupled with a robust system featuring an AMD Ryzen 9 5900HX processor, 16GB RAM (3200 MT/s), and running on Microsoft Windows 11, enabled efficient parallel processing, accelerating model training and facilitating rapid experimentation with various hyperparameter configurations. The system's memory capacity played a vital role in meeting the computational requirements of the proposed model, particularly for processing high-resolution brain MRI scans, allowing for efficient and uninterrupted training. The training was carried out using TensorFlow version 2.17.1, which offered a stable and adaptable environment for conducting transfer learning experiments.

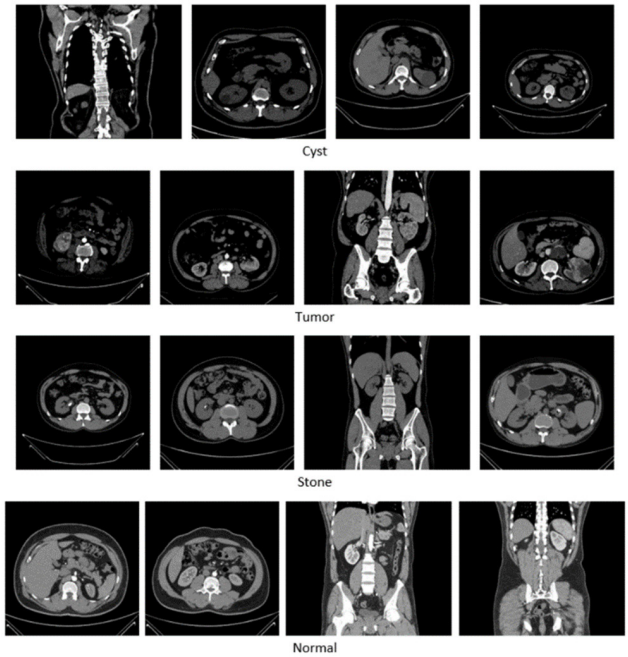


FIGURE 3. The visual representation of sample images within the KD dataset.

A. DATASETS

This section provides a detailed description of the datasets utilized in this study, namely KD, AKD, KD – D, and AD – D.

1) KD DATASET

The foundation of this study rests upon the KD dataset, a baseline collection obtained from [27] as detailed in Section "Original (Public) Dataset." Figure 3 provides a visual representation of sample images within this dataset. The KD dataset encompasses four distinct classes: cysts, tumors, stones, and normal kidney images, serving as the cornerstone for our subsequent analyses.

2) AKD DATASET

The AKD dataset was generated through the application of a diverse array of 11 augmentation techniques

on the KD dataset. The AutoAlbum strategy, outlined in the Section "Augmented Dataset," guided the selection and application of these augmentations. For each chosen technique, the corresponding hyperparameters were carefully tuned by using a grid search hyperparameter tuning algorithm [34].

Noise augmentations were applied to simulate various image acquisition artifacts. Gaussian noise was introduced with a mean of 0 and a standard deviation of 0.37 to replicate common distortions in image capture. Poisson noise was applied with a factor of 1.4, mimicking variations in photon count during image acquisition, which is typical in low-light conditions. Speckle noise was added with a variance of 0.25 to simulate variations in image intensity caused by interference patterns. Additionally, salt-and-pepper noise was introduced with an amount of 0.006, representing random pixel errors that can occur due to transmission or sensor imperfections.

Geometric transformations were applied to augment the dataset by altering the orientation of images. A horizontal flip was used to mirror the images along the vertical axis, creating a left-to-right inversion. A vertical flip mirrored the images along the horizontal axis, flipping them top to bottom. Additionally, a combination of both horizontal and vertical flips was applied, resulting in an inversion of the images along both axes.

Intensity augmentations were applied to introduce variations in image brightness and contrast. Random brightness and contrast adjustments were made within a range of $\pm 20\%$, altering the overall appearance of the images to simulate different lighting conditions. Additionally, random gamma correction was applied, with gamma values varying between 75% and 150% of the original intensity.

Blurring augmentations were used to simulate common image distortions caused by various factors. Gaussian blur was applied with a kernel size randomly selected between 1 and 3 pixels, softening the image and simulating the effects of camera focus issues or slight motion during capture. Motion blur was introduced with a kernel size of 2 pixels to mimic the artifacts caused by movement during image acquisition, such as when the camera or subject is in motion.

These ranges ensured that the introduced variations were both realistic and preserved the crucial diagnostic features of the kidney diseases within the images. This approach resulted in an augmented dataset that effectively simulates the inherent variability encountered in real-world clinical settings, thereby enhancing the robustness and generalizability of the subsequent models. Figure 4 provides a visual representation of sample images within this dataset.

3) KD–D DATASET

The KD – D dataset was derived from the original KD dataset, following the distributed approach outlined in Section "Public Distributed Dataset." Figure 5 provides a visual representation of sample images from four representative clients within this distributed dataset, showcasing the inherent diversity and heterogeneity that characterize real-world clinical data.

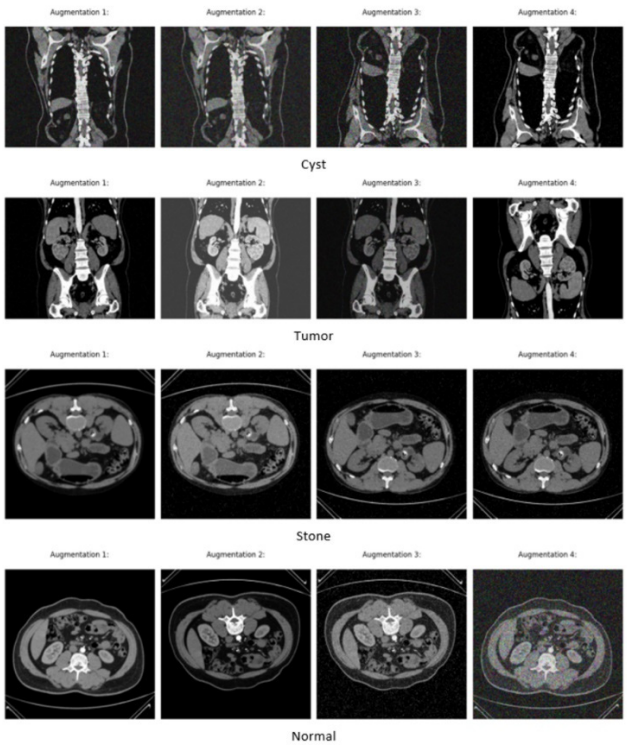


FIGURE 4. The visual representation of sample images within the AKD dataset.

4) AD–D DATASET

The AD – D dataset was generated from the KD – D dataset, as detailed in Section "Augmented Distributed Dataset." This crucial step involved applying the AutoAlbum augmentation strategy, incorporating a diverse set of 11 augmentation techniques with carefully selected hyperparameters as outlined in Section "AKD Dataset." Figure 6 showcases representative images from four distinct clients (used in this study) within the AD – D dataset, providing a visual glimpse into the diverse and augmented data utilized in this study.

B. TRAINING SETUP

In order to thoroughly assess the effectiveness of the proposed model, the datasets were partitioned into training, validation, and testing sets. The training set comprised 70% of the data, providing a substantial foundation for model learning. The validation set, constituting 20% of the data, served as a crucial checkpoint for monitoring model performance during the training process, enabling hyperparameter tuning and early stopping to prevent overfitting. The remaining 10% of the data was reserved for the final evaluation of the trained model's generalization capabilities on unseen data. Table 1 presents an overview of how the data is distributed across KD and AKD datasets in a centralized manner, facilitating a comprehensive understanding of the evaluation framework.

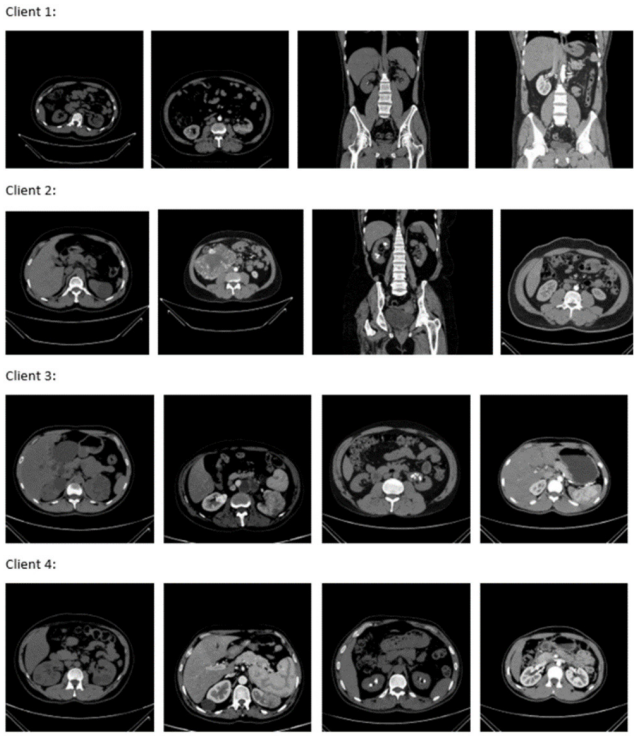


FIGURE 5. The visual representation of sample images within the KD – D dataset; 1st column: Cyst, 2nd column: Tumor, 3rd column: Stone, Last column: Normal.

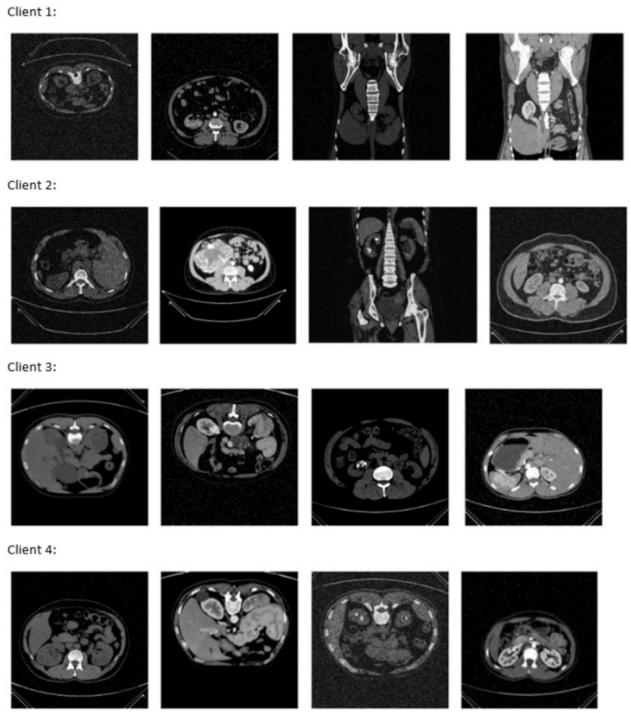


FIGURE 6. The visual representation of sample images within the AD – D dataset; 1st column: Cyst, 2nd column: Tumor, 3rd column: Stone, Last column: Normal.

For the KD – D dataset, the original KD dataset was distributed among four clients. Following a 70-20-10 split, a total of 8736 images were allocated for training, 2465 for validation, and 1245 for testing, constituting 70%, 20%, and 10% of the total dataset (12,446 images), respectively. The AD-D dataset comprises a total of 24,892 images, resulting from the augmentation and addition of new images to the original KD dataset. 17,472 images (approximately 70%) were allocated for training, 4,930 images (approximately 20%) were reserved for validation, and 2,490 images (approximately 10%) were designated for testing. Crucially, the test sets of these two datasets remained a unified set. This unified test set is essential for evaluating the efficacy of the global model after averaging the parameters received from all four clients, providing an unbiased assessment of the model's generalization performance on unseen data. Detailed statistics of these data distributions are presented in Table 2.

TABLE 1. The overview of the data distribution across KD and AKD datasets.

Dataset	Class Name	Training Set	Validation Set	Test Set
KD	Cyst	2596	742	371
	Stone	964	275	138
	Tumor	1598	457	228
	Normal	3554	1015	508
AKD	All Classes	3500	1000	500

TABLE 2. Detailed statistics of KD – D and AD – D data distribution.

Dataset	Class	Train (Split1, Split1, Split3, Split4)	Validation (Split1, Split1, Split3, Split4)	Test
KD – D	Cyst	612, 679, 633, 679	175, 192, 183, 185	371
	Stone	239, 253, 229, 245	69, 68, 72, 64	138
	Tumor	439, 358, 406, 400	121, 98, 117, 116	228
	Normal	894, 894, 916, 860	251, 258, 244, 252	508
AD – D	Cyst	1291, 1291, 1312, 1312	360, 367, 375, 368	742
	Stone	484, 492, 482, 474	133, 137, 140, 136	276
	Tumor	839, 797, 764, 806	237, 219, 215, 233	456
	Normal	1754, 1788, 1810, 1776	503, 509, 502, 496	1016

C. OPTIMIZING MODEL PERFORMANCE: HYPERPARAMETER TUNING AND EARLY STOPPING

To mitigate the risk of overfitting and maximize the performance of our deep learning models for kidney disease classification, a grid search hyperparameter tuning strategy was implemented [34]. Recognizing the potential for overfitting, where models may inadvertently learn spurious correlations within the training data that do not generalize

well to unseen cases, we systematically explored various combinations of batch sizes (32, 64, and128) and learning rates (0.0001, 0.001, 0.01, and 0.1) across the 21 PT-DCNN models under consideration. This exhaustive search aimed to identify the optimal hyperparameter configurations that would minimize overfitting and maximize model generalization. To further enhance model robustness, EarlyStopping was incorporated into the training process [35]. This method tracks the model's validation performance and stops the training process if there is no improvement in the validation loss over a specified number of consecutive epochs. By preventing excessive training, EarlyStopping effectively mitigates overfitting and encourages the model to learn more generalizable features from the training data. The hyperparameter tuning process was conducted using the AKD dataset, which offers a well-balanced distribution of classes and a wide range of image variations, ensuring a reliable and comprehensive evaluation. The optimal hyperparameter settings, including a learning rate of 0.001 and a batch size of 32, were determined for each model. The Adam optimizer and categorical cross-entropy were used in this study for each PT-DCNN model. The optimized epoch of each PT-DCNN used in this study is summarized in Table 3.

TABLE 3. The optimized epoch of 21 PT-DCNN used in this study.

PT-DCNN	Optimized Epoch	PT-DCNN	Optimized Epoch	PT-DCNN	Optimized Epoch
V19	8	EB4	10	D201	11
IV1	9	EB5	7	R50	11
IV3	11	EB6	9	R101	10
EB0	11	EB7	7	R152	5
EB1	10	D121	9	MV2	14
EB2	7	D161	12	MV3S	12
EB3	8	D169	14	MV3L	9

*V19: VGG19, IV1: InceptionV1, IV3: InceptionV3, EB (0 – 7): EfficientNetB (0 – 7), D (121 – 201): DenseNet (121 – 201), R (50 – 152): ResNet (50 – 152), MV2: MobileNet V2, MV3S: MobileNetV3 Small, MV3L: MobileNetV3 Large.

D. CUSTOM PT-DCNN AS THE GLOBAL MODEL FOR THE FEDERATED LEARNING

To determine the most suitable backbone for our federated learning framework, we conducted a thorough evaluation of 21 PT-DCNN architectures. This analysis was performed using the AKD dataset, known for its balanced class representation and diverse imaging characteristics, ensuring a fair and robust comparison. Each architecture was assessed based on key metrics such as classification accuracy and training duration. The outcomes of this comparative study are presented in Table 4. This detailed evaluation informed the selection of the most effective PT-DCNN model, laying a strong foundation for further optimization and integration into the final federated learning system aimed at kidney disease classification.

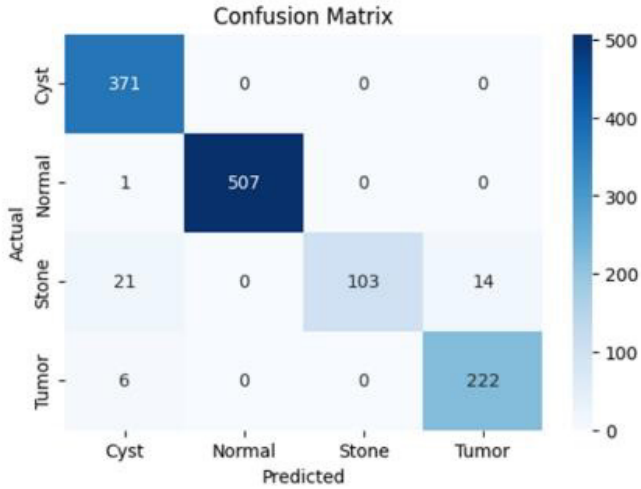
The choice of EfficientNetB0 as the base or global model was driven by a comprehensive evaluation of 21 distinct

TABLE 4. The optimized epoch of 21 PT-DCNN used in this study.

PT-DCNN	Test Accuracy, Train Time	PT-DCNN	Test Accuracy, Train Time	PT-DCNN	Test Accuracy, Train Time
V19	0.76, 8 hrs	EB4	0.57, 6 hrs	D201	0.70, 27 mins
IV1	0.79, 1 hr	EB5	0.61, 14 hrs	R50	0.63, 15 mins
IV3	0.68, 4 hr	EB6	0.55, 17 hrs	R101	0.66, 16 mins
EB0	0.91, 38 mins	EB7	0.62, 26 hrs	R152	0.63, 34 mins
EB1	0.88, 50 mins	D121	0.89, 23 mins	MV2	0.79, 12 mins
EB2	0.79, 1 hr	D161	0.86, 1 hr	MV3S	0.80, 8 mins
EB3	0.60, 3 hr	D169	0.81, 18 mins	MV3L	0.77, 11 mins

PT-DCNN architectures, as summarized in the provided table. While models like DenseNet121 and DenseNet161 demonstrated competitive performance with higher test accuracies, DenseNet161's training times were significantly longer compared to EfficientNetB0. This extended training time could pose challenges in the context of federated learning, where communication overhead and training time at each client significantly impact the overall training efficiency. EfficientNetB0, despite exhibiting slightly higher training time than models like DenseNet121, demonstrated a remarkable balance between accuracy and training efficiency. Its high test accuracy of 0.91, coupled with a relatively moderate training time of 38 minutes, indicated strong performance with acceptable computational cost. This combination of high accuracy and reasonable training time made EfficientNetB0 the most suitable choice as the base model for our federated learning framework, ensuring both effective model performance and efficient training across the distributed network of clients.

To tailor the pre-trained EfficientNetB0 model for our specific kidney disease classification task, a grid search hyperparameter tuning process was conducted. This focused on optimizing the classification head, which is appended to the base EfficientNetB0 model. Six distinct configurations were explored, each comprising a varying number of densely connected layers with different numbers of neurons. These configurations included: (64), (128), (256, 128), (512, 256, 128), (512, 256, 128, 64), and (1024, 512, 128, 64), where each set of numbers within parentheses represents the number of neurons in each successive dense layer. For instance, the configuration (512, 256, 128) denotes a classification head with three dense layers containing 512, 256, and 128 neurons, respectively. Through rigorous experimentation, the optimal classification head architecture was determined to be (256, 128), followed by a final output layer with four neurons, corresponding to the four distinct classes within our dataset: cyst, tumor, stone, and normal. This optimized architecture ensures



Loss	Accuracy	Precision	Recall	F1 Score
0.0033	0.9663	0.9678	0.9663	0.9646

FIGURE 7. The visual representation of the model's performance at the culmination of the 5th training round on KD – D dataset.

that the model effectively extracts relevant features from the EfficientNetB0 backbone while simultaneously achieving high classification accuracy for kidney disease detection. Overall this model is named custom EfficientNetB0 which was used as the global model in this study.

E. CUSTOM EFFICIENTNETB0 MODEL EVALUATION ON THE KD-D DATASET

Following the training and validation of the custom EfficientNetB0 model on the KD – D dataset, the federated learning process, utilizing the FedAvg algorithm, was employed to refine the model parameters. Subsequently, the performance of the refined model was rigorously evaluated on the held-out test set of the KD – D dataset. While the federated learning training process was conducted over 10 rounds of communication and parameter updates, it was observed that the performance metrics stabilized after the 5th round. No significant improvements were observed beyond the 5th round. Therefore, this study focuses on reporting the performance metrics up to the 5th round, providing a comprehensive overview of the model's learning trajectory and demonstrating the point of diminishing returns in the federated learning process. Figure 7 provides a visual representation of the model's performance at the culmination of the 5th training round, showcasing the confusion matrix for a comprehensive understanding of its classification accuracy. To provide a holistic view of the model's performance throughout the training process, in Figure 7 we included the key performance metrics, including loss, accuracy, precision, recall, and F1-score, of the 5th training rounds. The entire federated learning process, encompassing 5 rounds of communication and model updates, was completed within approximately 7 minutes. This evaluation provides valuable insights into the model's ability to accurately classify kidney diseases, including cysts, tumors, and stones, while also identifying potential areas for further improvement.

The model achieved an impressive overall accuracy of 0.9663 on the KD – D dataset, indicating a high level of agreement between the model's predictions and the actual ground truth labels. This strong performance is further corroborated by the high precision (0.9678) and recall (0.9663) values, suggesting that the model effectively identifies true positive cases while minimizing false positives and false negatives. The F1-score of 0.9646, which represents the harmonic mean of precision and recall, further reinforces the model's overall performance, demonstrating a robust balance between these two critical metrics. The low loss value of 0.0033 further indicates that the model has effectively learned the underlying patterns within the data and minimized the discrepancy between its predictions and the true labels. These results collectively demonstrate the exceptional performance of the custom EfficientNetB0 model on the KD – D dataset, highlighting its potential for accurate and reliable kidney disease classification in a distributed environment using federated learning.

We used Grad-CAM visualizations on correctly classified samples from the KD–D dataset (Figure 8), showing one example from each of the four classes. The heatmaps highlight the key image regions the model used for classification, confirming it focuses on clinically relevant features. These consistent, class-specific patterns suggest the model generalizes well across the federated learning setup, even with distributed and imbalanced data.

F. CUSTOM EFFICIENTNETB0 MODEL EVALUATION ON THE AD-D DATASET

To further evaluate the robustness and generalizability of the trained model, its performance was assessed on the

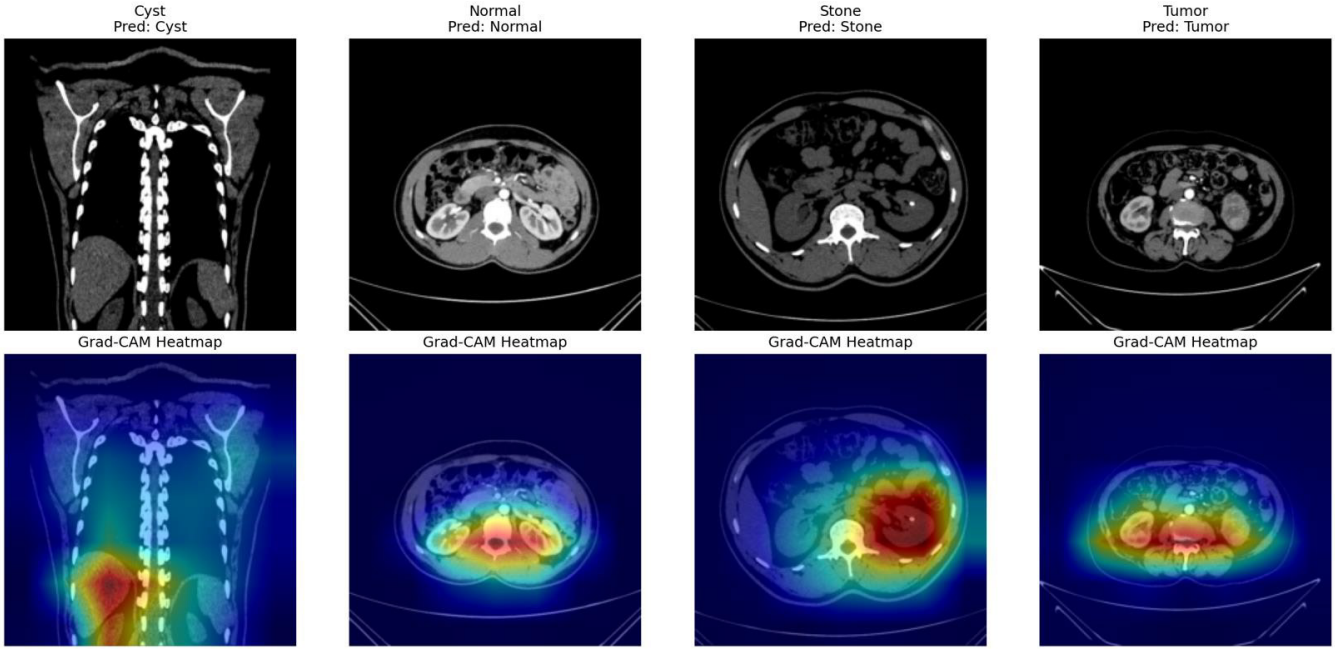


FIGURE 8. Grad-CAM visualizations of correctly classified samples from the KD–D dataset, one from each of the four classes.

augmented and distributed AD – D dataset. Following a federated learning process comprising 10 rounds of communication and model updates, the model's performance was evaluated on the held-out test set of the AD – D dataset. Figure 9 provides a comprehensive visualization of the model's performance on this challenging dataset (after 5th round), presenting the confusion matrix and key performance metrics such as accuracy, precision, recall, and F1-score. This evaluation of the augmented and distributed data provides valuable insights into the model's ability to effectively handle the complexities and variations within the federated learning framework.

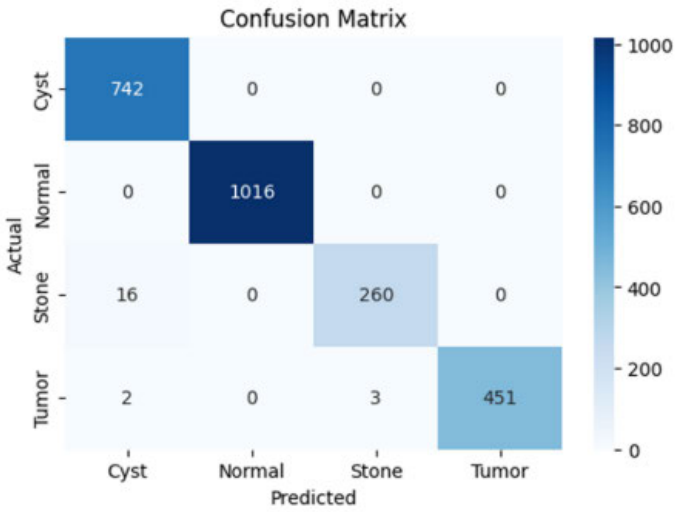
The model achieved exceptional performance on the AD – D dataset, achieving an accuracy of 0.9916 with a very low loss of 0.0008. High precision (0.9917) and recall (0.9916) values further demonstrate the model's ability to accurately identify true positive cases while minimizing false positives and negatives. The F1-score of 0.9915 further reinforces this strong performance, indicating a well-balanced model with high predictive accuracy. These results demonstrate the robustness and adaptability of the custom EfficientNetB0 model, even when trained and evaluated on the challenging and augmented AD – D dataset, showcasing its potential for effective and reliable kidney disease classification in real-world clinical settings.

To further enhance the interpretability of the central model's predictions, we incorporated Grad-CAM visualizations on correctly identified samples from the AD–D dataset (Figure 10). Specifically, we present heatmaps for one representative sample from each of the four classes, highlighting the discriminative regions that the model focused on during classification. These visualizations offer insight into how the model distinguishes between different types of kidney disease and validate that it is attending to clinically relevant regions in the input images. The consistent and class-specific activation patterns observed across these samples demonstrate the model's ability to generalize well within the federated learning framework, even when trained on distributed, augmented datasets.

G. PERFORMANCE COMPARISON OF PT-DCNN MODELS WITH THE PROPOSED CUSTOM EFFICIENTNETB0

This section delves into a comparative analysis of the custom EfficientNetB0 model's performance with the performance of 20 distinct PT-DCNN architectures used in this study on both the KD – D and AD – D datasets. This comparative analysis aims to provide valuable insights into the relative performance of these architectures in the context of distributed and augmented data, simulating the challenges of real-world federated learning scenarios. By evaluating these models on both datasets, we can gain a deeper understanding of their strengths and weaknesses in handling data heterogeneity and variations. Furthermore, this comparative analysis will serve as a benchmark to assess the performance of our custom EfficientNetB0 model, highlighting its strengths and identifying potential areas for further improvement within the broader landscape of pre-trained architectures for kidney disease classification. Table 5 presents the results obtained from the analysis.

The analysis reveals a significant performance advantage for the custom EfficientNetB0 model, particularly on the more challenging AD – D dataset. While models like D121



Loss	Accuracy	Precision	Recall	F1 Score
0.0008	0.9916	0.9917	0.9916	0.9915

FIGURE 9. The visual representation of the model's performance at the culmination of the 5th training round on AD – D dataset.

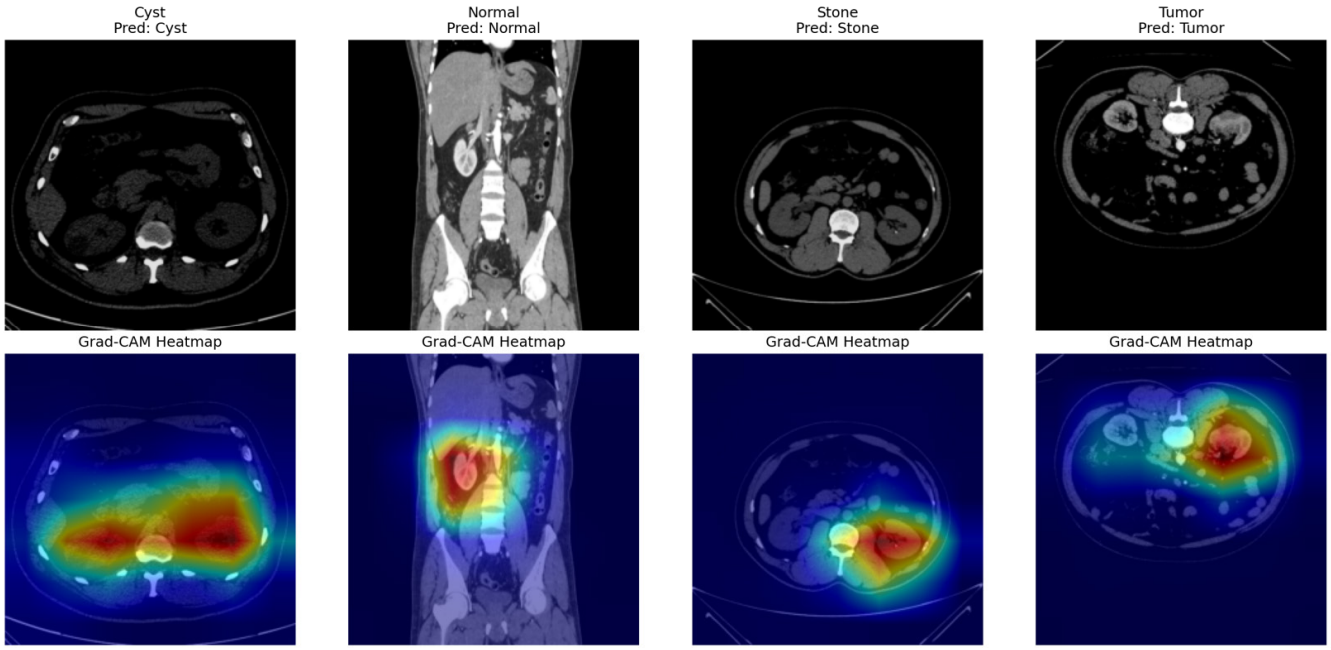


FIGURE 10. Grad-CAM visualizations of correctly classified samples from the AD–D dataset, one from each of the four classes.

and D161 demonstrated strong performance on the AD – D dataset, their accuracy decreased noticeably on the KD – D dataset, highlighting the impact of data augmentation and distribution on their generalization capabilities. In contrast, the custom EfficientNetB0 achieved a remarkable accuracy of 0.9916 on the AD – D dataset, significantly surpassing all other models. This superior performance underscores the effectiveness of the proposed federated learning framework and the robustness of the custom EfficientNetB0 model in adapting to the complexities of the augmented and distributed data, demonstrating its potential for accurate and reliable kidney disease classification in diverse and challenging clinical scenarios.

V. DISCUSSION

Early and accurate diagnosis of kidney disease is crucial for effective treatment and management, significantly improving patient outcomes and potentially preventing the progression

TABLE 5. The comparative analysis of the custom EfficientNetB0 model's performance with the performance of 20 distinct PT-DCNN architectures used in this study.

Model	Accuracy on KD – D Dataset	Accuracy on AD – D dataset	Model	Accuracy on KD – D Dataset	Accuracy on AD – D dataset
V19	0.71	0.79	D161	0.85	0.89
IV1	0.75	0.76	D169	0.79	0.82
IV3	0.66	0.69	D201	0.70	0.75
EB1	0.82	0.85	R50	0.62	0.67
EB2	0.77	0.78	R101	0.67	0.69
EB3	0.55	0.62	R152	0.62	0.66
EB4	0.54	0.57	MV2	0.75	0.82
EB5	0.60	0.63	MV3S	0.83	0.84
EB6	0.56	0.57	MV3L	0.76	0.79
EB7	0.60	0.63	Proposed Custom EB0	0.97	0.99
D121	0.86	0.91			

to end-stage renal disease. Traditional diagnostic methods can be time-consuming, labor-intensive, and may require specialized expertise. Deep learning offers a promising solution by using its ability to automatically learn complex patterns and features from medical images, such as ultrasounds or CT scans. Federated learning offers a promising solution by enabling collaborative model training across multiple institutions while preserving data privacy. In the context of kidney disease detection, federated learning allows healthcare institutions to jointly train a high-performing model without sharing their sensitive patient data. This decentralized approach addresses critical challenges such as data privacy concerns, data silos, and resource limitations, which can hinder the development and deployment of effective AI-powered diagnostic tools. By using federated learning, healthcare institutions can collaboratively learn from diverse and geographically distributed datasets, improving the model's generalizability and robustness. This collaborative approach can lead to more accurate and equitable access to high-quality diagnostic tools, ultimately improving patient outcomes and enhancing the overall quality of healthcare. These were the main motivations for using federated learning in this study.

This study investigates the efficacy of a federated learning framework for kidney disease detection, employing a custom EfficientNetB0 model as the base architecture. In our study, we used four clients to simulate a realistic federated learning scenario for decentralized kidney disease detection. Our choice was motivated by several practical considerations and the need to balance the complexity of the simulation with the ability to draw meaningful conclusions: (1) Computational Resources: Conducting extensive experiments with a very large number of clients can be computationally expensive and time-consuming. We aimed to strike a balance by using a sufficient number of clients to demonstrate the feasibility of our approach while keeping the computational demands manageable. (2) Foundation of Concept: This study serves as a proof of concept for our proposed federated learning framework for kidney disease detection. While we acknowledge that real-world deployments may involve a larger number of clients, our experiments with four clients provide valuable insights into the framework's effectiveness and its ability to handle data imbalance. (3) Clarity of Results: Using a smaller number of clients can sometimes make it easier to analyze and interpret the results of our experiments. With fewer clients, we can more clearly attribute performance differences to specific factors, such as the presence or absence of data augmentation. While we recognize that the number of clients can influence the model's performance, our experiments with four clients provide a valuable foundation for future research in this area.

The study specifically aimed to evaluate the framework's robustness by training and evaluating the model on both the high-quality KD – D dataset and the more challenging AD – D dataset, which incorporates diverse and noisy augmentations. This approach sought to emulate the real-world scenario where data quality and characteristics received from different hospitals may vary significantly. The performance of the federated learning model was assessed over five rounds of communication and parameter updates, with the results summarized in Table 6. Importantly, Table 6 presents the global model's performance metrics after training on both the KD – D and AD – D datasets after each round of federated learning. This comprehensive evaluation provides valuable insights into the framework's ability to effectively learn from diverse and potentially noisy data, demonstrating its potential for robust and reliable kidney disease detection in real-world clinical settings.

TABLE 6. The global model's performance metrics after training on both the KD – D and AD – D datasets after each round of federated learning.

D	RO	L	A	P	RE	F
KD – D	1	0.0052	0.9438	0.9451	0.9425	0.9438
	2	0.0045	0.9512	0.9528	0.9500	0.9514
	3	0.0039	0.9587	0.9595	0.9580	0.9588
	4	0.0036	0.9621	0.9632	0.9615	0.9624
	5	0.0033	0.9663	0.9678	0.9663	0.9646
AD – D	1	0.0015	0.9880	0.9885	0.9875	0.9880
	2	0.0012	0.9900	0.9905	0.9895	0.9900
	3	0.0010	0.9908	0.9915	0.9905	0.9910
	4	0.0009	0.9914	0.9916	0.9912	0.9914
	5	0.0008	0.9916	0.9917	0.9916	0.9915

*D: Dataset, RO: Round, L: Loss, A: Accuracy, P: Precision, RE: Recall, F: F1 Score

The federated learning framework demonstrated consistent and significant performance improvements across both the KD – D and AD – D datasets throughout the initial five rounds of training. On the KD – D dataset, the model exhibited a steady increase in accuracy, precision, recall, and F1-score, converging towards high-performance levels.

Notably, the model achieved a high accuracy of 0.9663 on the KD – D dataset in the 5th round, indicating its ability to effectively learn and generalize from the distributed data. On the more challenging AD – D dataset, characterized by its augmented and distributed nature, the model exhibited remarkable robustness and adaptability. The performance metrics on the AD – D dataset also demonstrated a consistent upward trend across the five rounds, culminating in an exceptional accuracy of 0.9916 in the 5th round. This significant performance on the AD – D dataset highlights the effectiveness of the federated learning framework in handling data heterogeneity and variations, showcasing its potential for robust and accurate kidney disease classification in real-world clinical settings.

When assessing deep learning models for kidney disease detection and classification, it's important to look beyond the performance metrics listed in Table 6. A thorough evaluation should also account for the model's architectural complexity—indicated by the number of trainable parameters—as well as its computational efficiency, including training duration and hardware resource usage. These factors are critical for determining the model's feasibility for real-world clinical deployment. A computationally expensive model may not be practical for routine clinical use, especially in resource-constrained settings. Table 7 presents an evaluation of these essential aspects for the proposed model, highlighting its overall efficiency and suitability for potential implementation in clinical settings for kidney disease diagnosis.

TABLE 7. The critical metrics for determining the model's feasibility for real-world clinical deployment.

Dataset	Training time (mins)	Testing time (secs)	Trainable Parameters	Computational resources
KD – D	7	30	361,348	T4 GPU High RAM
AD – D	12	50		

The provided data reveals key insights into the computational efficiency of the model. Both the KD – D and AD – D datasets were trained on a T4 GPU with high RAM, indicating a robust computational environment. The KD – D dataset exhibited a shorter training time of 7 minutes compared to 12 minutes for the AD – D dataset. This difference likely stems from the increased complexity of the augmented data in the AD – D dataset, requiring more iterations and computational resources for the model to converge. Consistent with expectations, the testing time was significantly shorter than the training time for both datasets. The KD – D dataset had a testing time of 30 seconds, while the AD – D dataset required slightly longer at 50 seconds. This disparity could be attributed to the increased size and complexity of the augmented data in the AD – D dataset, requiring more processing time for inference. Notably, the number of trainable parameters remained constant at 361,348 for both datasets. This indicates that the model's architecture and complexity were consistent across both training scenarios, with the observed differences in training and testing times primarily attributed to the characteristics of the datasets themselves. This analysis demonstrates a reasonable balance between training and testing times, suggesting that the model can be deployed efficiently in a clinical setting with acceptable computational requirements.

To further analyze the convergence behavior and robustness of the federated learning environment, we have included additional visualizations in the form of learning curves. Specifically, we present the plots of accuracy – loss vs. round (Figure 11),precision –recall vs. round (Figure 12) for both the KD–D and AD–D datasets. These metrics are tracked up to the 10th communication round, providing insight into the central model's performance progression during early-stage training. These visualizations help illustrate the stability, generalization, and learning dynamics of the model under federated settings.

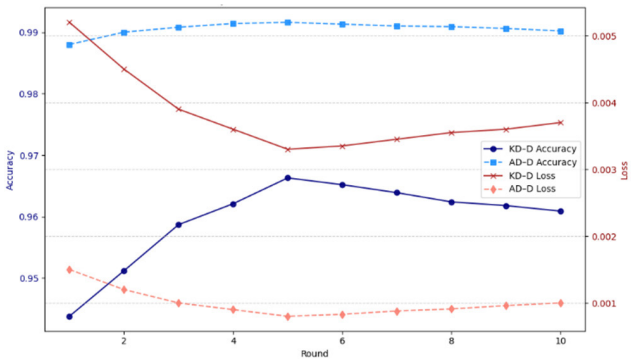


FIGURE 11. Accuracy and loss vs. round for the KD–D and AD–D datasets.

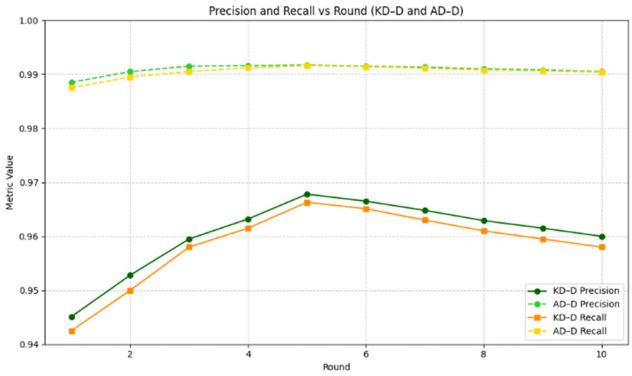


FIGURE 12. Precision and recall vs. round for the KD–D and AD–D datasets.

In Figure 11, the accuracy for both datasets shows a consistent upward trend during the first five rounds, indicating steady model improvement in the early stages of training. The AD–D dataset achieves notably higher accuracy compared to KD–D, demonstrating the benefit of augmented data diversity introduced through the AutoAlbum strategy. After round 5, a slight decline or plateau in accuracy is observed for both

datasets, suggesting the onset of convergence. This behavior is typical in federated settings where client data heterogeneity and limited local updates can lead to early saturation. Correspondingly, the loss values decrease steadily until round 5 and then begin to stabilize or slightly increase, indicating that the model may be approaching its optimal generalization capacity.

Figure 12 further supports this observation by showing the evolution of precision and recall across communication rounds. Both metrics exhibit high and stable values throughout the rounds, with AD–D consistently outperforming KD–D. The AD–D model maintains a precision and recall above 0.99 in the early rounds, reflecting its improved ability to distinguish between classes. After round 5, a slight downward trend is observed in both precision and recall, more noticeably in KD–D, possibly due to overfitting on non-diverse client data or local model drift. Despite this minor decline, the overall performance remains robust, reinforcing the model's effectiveness in a decentralized learning environment. These plots collectively demonstrate that the federated learning system not only converges within a small number of communication rounds but also maintains stable performance even in the presence of data heterogeneity, particularly when enhanced through augmentation strategies like AutoAlbum.

Table 8 presents a comparative analysis of the proposed federated learning model with other state-of-the-art approaches reported in the literature for kidney disease detection and classification. This comparative analysis provides valuable insights into the relative performance of the proposed model within the broader context of existing research. By comparing the precision achieved by these different approaches on both the KD – D and augmented AD – D test datasets, we can objectively assess the strengths and weaknesses of the proposed model and identify areas for potential future improvements. Given that both the KD – D and AD – D datasets exhibit class imbalance, precision was selected as the primary performance metric for the comparative analysis. In imbalanced datasets, where one class significantly outnumbers others, accuracy can be misleading as it may be inflated by the majority class. Precision, on the other hand, focuses on the quality of positive predictions, measuring the proportion of correctly predicted positive instances among all predicted positive instances. In the context of kidney disease detection, false positives can have significant consequences, such as unnecessary invasive procedures or anxiety for patients. By prioritizing precision, the evaluation emphasizes the model's ability to minimize false positive predictions, ensuring that positive diagnoses are reliable and minimizing the risk of unnecessary interventions.

Table 8 highlights a significant performance drop for many existing approaches on the augmented AD – D dataset compared to the KD – D dataset, particularly those involving segmentation techniques. This observation underscores the challenges posed by augmented data, which introduces variations in image appearance that can significantly impact the performance of segmentation-based methods. These methods may struggle to accurately segment regions of interest in augmented images, leading to erroneous feature extraction and ultimately degrading classification performance. This observation highlights the importance of robustness to data augmentation in real-world clinical settings. The proposed federated learning approach, utilizing the custom EfficientNetB0 model, demonstrates exceptional robustness, achieving a precision of 0.9678 on the KD – D dataset and a remarkable 0.9917 on the challenging AD – D dataset. This significant outperformance underscores the effectiveness of the federated learning framework and the robustness of the custom EfficientNetB0 model in handling the complexities introduced by data augmentation and distribution. This analysis emphasizes the superior performance of the proposed model compared to existing approaches, particularly in handling the challenges presented by the augmented AD – D dataset. This highlights the potential of the proposed federated learning framework for robust and accurate kidney disease classification in diverse and challenging real-world clinical settings.

TABLE 8. The comparative analysis of the proposed federated learning model with other state-of-the-art approaches reported in the literature for kidney disease detection and classification.

Methodology Used	Precision on the KD – D Dataset	Precision on the AD – D Dataset
The methodology used in [12]	0.9041	0.7367
The methodology used in [13]	0.9503	0.9653
The methodology used in [14]	0.9700	0.9769
The methodology used in [15]	0.9324	0.8744
The methodology used in [16]	0.9467	0.9652
The methodology used in [17]	0.8455	0.7148
The methodology used in [18]	0.9012	0.8256
The methodology used in [19]	0.9431	0.8521
The methodology used in [20]	0.8589	0.6893
The methodology used in [21]	0.9476	0.9687
The methodology used in [22]	0.9124	0.8738
The methodology used in [23]	0.6235	0.5641
The methodology used in [24]	0.9358	0.8137
The methodology used in [25]	0.7463	0.5962
The methodology used in [26]	0.9462	0.9539
The Proposed Approach	0.9678	0.9917

VI. CONCLUSION

This study presents a new approach to kidney disease detection and classification utilizing a federated learning framework. This framework uses the power of deep learning, specifically a custom EfficientNetB0 model, to effectively identify subtle patterns and characteristics that distinguish between healthy kidney tissue and various disease conditions such as cysts, tumors, and stones.

This study addresses the challenges of Non-IID data and data asymmetry through AutoAlbum, an automated augmentation technique that introduces realistic variations into the dataset using techniques such as Poisson Noise, Gaussian Noise, Speckle Noise, Salt-and-Pepper Noise, various Flips, Random Brightness & Contrast, Random Gamma,

Gaussian Blur, and Motion Blur, enhancing model robustness and generalization. The creation of KD – D and AD – D datasets simulates real-world federated learning scenarios, enabling rigorous evaluation of the model's performance in handling Non-IID data distributions. The proposed framework demonstrates robust performance in a decentralized setting, achieving results that are competitive with centralized approaches while upholding patient privacy.

In addition to the strong quantitative results, Grad-CAM visualizations further support the effectiveness of the proposed framework by offering visual explanations of the model's decision-making process. These heatmaps, generated for correctly classified samples across all kidney disease classes, reveal that the model consistently focuses on clinically relevant regions within the images. This not only validates the model's interpretability but also reinforces trust in its predictions, especially in a healthcare setting where understanding model rationale is crucial. The consistent and class-specific activation patterns observed through Grad-CAM highlight the model's ability to generalize effectively across distributed, augmented data within a federated learning environment.

While the proposed federated learning framework demonstrates promising results in kidney disease detection, several avenues for future improvement can be explored. Firstly, the current study primarily focuses on classifying kidney conditions (tumor, cyst, normal, stone). Expanding the scope to include tasks such as lesion segmentation and localization within the kidney would provide more comprehensive diagnostic information, aiding in treatment planning and surgical guidance. Secondly, while the framework has demonstrated robustness to data variations within the current study, further investigation is needed to assess its performance in scenarios with even more extreme data heterogeneity, such as significant variations in image acquisition protocols, patient demographics, and disease prevalence across participating institutions. This will provide valuable insights into the framework's adaptability and robustness in real-world clinical settings with diverse and challenging data distributions. Thirdly, the current study has used four clients. Further research is needed to investigate the optimal number of clients and the impact of varying client participation rates in the context of kidney disease detection. Finally, exploring the integration of other explainable AI techniques, such as attention mechanisms and feature visualization, can enhance the interpretability of the model's predictions, fostering greater trust and facilitating clinical adoption.

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REFERENCES

[1] *Polycystic Kidney Disease*. Accessed: Aug. 1, 2023. [Online]. Available: https://www.kidney.org/kidney-topics/polycystic-kidney-disease

[2] *Key Statistics About Kidney Cance*. Accessed: Aug. 1, 2023. [Online]. Available: https://www.cancer.org/cancer/types/kidney-cancer/about/key-statistics.html#:~:text=About%2081%2C610%20new%20cases%20of,been%20falling%20for%20many%20years

[3] *Kidney Stones*. Accessed: Aug. 1, 2023. [Online]. Available: https://www.kidney.org/kidney-topics/kidney-stones#::text=get%20kidney%20stones?-,How%20common%20are%20kidney%20stones?,What%20is%20a%20kidney%20stone?

[4] J. J. Grantham and V. E. Torres, "The importance of total kidney, volume in evaluating progression of polycystic kidney disease," *Nature Rev. Nephrol.*, vol. 12, no. 11, pp. 667–677, 2016.

[5] S. C. Campbell, P. E. Clark, S. S. Chang, J. A. Karam, L. Souter, and R. G. Uzzo, "Renal mass and localized renal cancer: Evaluation, management, and follow-up: AUA guideline: Part I," *J. Urol.*, vol. 206, no. 2, pp. 199–208, Aug. 2021.

[6] M. Sivaguru, J. J. Saw, E. M. Wilson, J. C. Lieske, A. E. Krambeck, J. C. Williams, M. F. Romero, K. W. Fouke, M. W. Curtis, J. L. Kear-Scott, N. Chia, and B. W. Fouke, "Human kidney stones: A natural record of universal biomineralization," *Nature Rev. Urol.*, vol. 18, no. 7, pp. 404–432, Jul. 2021.

[7] O. V. Prada, M. Á. Vera, G. Ramirez, R. B. Rojel, and D. M. Maldonado, "Statistical techniques for digital pre-processing of computed tomography medical images: A current review," *Displays*, vol. 85, Dec. 2024, Art. no. 102835.

[8] I. Alnazer, P. Bourdon, T. Urruty, O. Falou, M. Khalil, A. Shahin, and C. Fernandez-Maloigne, "Recent advances in medical image processing for the evaluation of chronic kidney disease," *Med. Image Anal.*, vol. 69, Apr. 2021, Art. no. 101960.

[9] M. Zhang, Z. Ye, E. Yuan, X. Lv, Y. Zhang, Y. Tan, C. Xia, J. Tang, J. Huang, and Z. Li, "Imaging-based deep learning in kidney diseases: Recent progress and future prospects," *Insights Imag.*, vol. 15, no. 1, p. 50, Feb. 2024.

[10] S. M. Ganie, P. K. D. Pramanik, and Z. Zhao, "Deep transfer learning for kidney disease detection using CT scan images," in *Proc. IEEE Int. Conf. Bioinf. Biomed. (BIBM)*, Dec. 2024, pp. 5624–5629.

[11] A. Rauniyar, D. H. Hagos, D. Jha, J. E. Håkegård, U. Bagci, D. B. Rawat, and V. Vlassov, "Federated learning for medical applications: A taxonomy, current trends, challenges, and future research directions," *IEEE Internet Things J.*, vol. 11, no. 5, pp. 7374–7398, Mar. 2024.

[12] Y. Kumar, T. P. S. Brar, C. Kaur, and C. Singh, "A comprehensive study of deep learning methods for kidney tumor, cyst, and stone diagnostics (don't short) and detection using CT images," *Arch. Comput. Methods Eng.*, vol. 2024, pp. 1–26, May 2024.

[13] S. Asif, M. Awais, and S. U. R. Khan, "IR-CNN: Inception residual network for detecting kidney abnormalities from CT images," *Netw. Model. Anal. Health Informat. Bioinf.*, vol. 12, no. 1, p. 35, Sep. 2023.

[14] M. N. Islam, M. Al Mamun, M. F. Faruk, A. Y. Srizon, S. M. M. Hasan, and B. Roy, "Spatial attention-guided deep learning for accurate kidney disease classification in CT scans," in *Proc. 26th Int. Conf. Comput. Inf. Technol. (ICCIT)*, Dec. 2023, pp. 1–6.

[15] A. Altalbe, "Correction: Applying customized convolutional neural network to kidney image volumes for kidney disease detection," *Comput. Syst. Sci. Eng.*, vol. 48, no. 4, pp. 1075–1081, 2024.

[16] A. Bhattacharjee, S. Rabea, A. Bhattacharjee, E. B. Elkaeed, R. Murugan, H. M. R. M. Selim, R. K. Sahu, G. A. Shazly, and M. M. Salem Bekhit, "A multi-class deep learning model for early lung cancer and chronic kidney disease detection using computed tomography images," *Frontiers Oncol.*, vol. 13, Jun. 2023, Art. no. 1193746.

[17] M. S. Hossain, S. M. N. Hassan, M. Al-Amin, M. N. Rahaman, R. Hossain, and M. I. Hossain, "Kidney disease detection from CT images using a customized CNN model and deep learning," in *Proc. Int. Conf. Adv. Intell. Comput. Appl. (AICAPS)*, Feb. 2023, pp. 1–6.

[18] F. Maqsood, W. Zhenfei, M. M. Ali, B. Qiu, N. U. Rehman, F. Sabah, T. Mahmood, I. Din, and R. Sarwar, "Artificial intelligence-based classification of CT images using a hybrid SpinalZFNet," *Interdiscipl. Sciences: Comput. Life Sci.*, vol. 16, no. 4, pp. 907–925, Dec. 2024.

[19] S. Asif, X. Zheng, and Y. Zhu, "An optimized fusion of deep learning models for kidney stone detection from CT images," *J. King Saud Univ.-Comput. Inf. Sci.*, vol. 36, no. 7, Sep. 2024, Art. no. 102130.

[20] E. Özbay, F. A. Özbay, and F. S. Gharehchopogh, "Kidney tumor classification on CT images using self-supervised learning," *Comput. Biol. Med.*, vol. 176, Jun. 2024, Art. no. 108554.

[21] N. Sasikaladevi, S. Pradeepa, A. Revathi, S. Vimal, and R. G. Crespo, "Diagnosis of kidney cyst, tumor and stone from CT scan imagesusing feature fusion hypergraph convolutional neural network (F2HCN2)," *Int. J. Multiscale Comput. Eng.*, vol. 22, no. 5, pp. 35–46, 2024.

[22] V. Karthikeyan, M. N. Kishore, and S. Sajin, "End-to-end light-weighted deep-learning model for abnormality classification in kidney CT images," *Int. J. Imag. Syst. Technol.*, vol. 34, no. 1, p. 23022, Jan. 2024.

[23] S. Asif, M. Zhao, X. Chen, and Y. Zhu, "StoneNet: An efficient lightweight model based on depthwise separable convolutions for kidney stone detection from CT images," *Interdiscipl. Sci., Comput. Life Sci.*, vol. 15, no. 4, pp. 633–652, Dec. 2023.

[24] J. Sarada, N. V. M. Lakshmi, and T. L. Praveena, "U-Capkidnets++-: A novel hybrid capsule networks with optimized deep feed forward networks for an effective classification of kidney tumours using CT kidney images," *Int. J. Recent Innov. Trends Comput. Commun.*, vol. 10, no. 1s, pp. 274–283, Dec. 2022.

[25] X. Fu, H. Liu, X. Bi, and G. Xiao, "Deep-learning-based CT imaging in the quantitative evaluation of chronic kidney diseases," *J. Healthcare Eng.*, vol. 2021, pp. 1–9, Jan. 2021.

[26] S. U. Rana, M. Nur-A-Alam, S. Akter, and M. N. H. Likhon, "Kidney-MultiNet: A web-based automatic system for kidney disease detection using hybrid machine learning model from CT scan images," *Biomed. Mater. Devices*, vol. 3, no. 2, pp. 885–896, Sep. 2025.

[27] *CT Kidney Dataset*. Accessed: Aug. 1, 2023. [Online]. Available: https://www.kaggle.com/datasets/nazmul0087/ct-kidney-dataset-normal-cyst-tumor-and-stone

[28] T. Sun, D. Li, and B. Wang, "Decentralized federated averaging," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 4, pp. 4289–4301, Apr. 2023.

[29] M. Moshawrab, M. Adda, A. Bouzouane, H. Ibrahim, and A. Raad, "Reviewing federated learning aggregation algorithms; strategies, contributions, limitations and future perspectives," *Electronics*, vol. 12, no. 10, p. 2287, May 2023.

[30] R. Azadnia, M. M. Al-Amidi, H. Mohammadi, M. A. Cifci, A. Daryab, and E. Cavallo, "An AI based approach for medicinal plant identification using deep CNN based on global average pooling," *Agronomy*, vol. 12, no. 11, p. 2723, Nov. 2022.

[31] Y. Shanmugarasa, H.-Y. Paik, S. S. Kanhere, and L. Zhu, "A systematic review of federated learning from clients' perspective: Challenges and solutions," *Artif. Intell. Rev.*, vol. 56, no. S2, pp. 1773–1827, Nov. 2023.

[32] J. Zhang, S. Guo, X. Ma, H. Wang, W. Xu, and F. Wu, "Parameterized knowledge transfer for personalized federated learning," in *Proc. Adv. Neural Inf. Process. Syst.*, 2021, pp. 10092–10104.

[33] B. J. Erickson and F. Kitamura, "Magician's corner: 9. performance metrics for machine learning models," *Radiol., Artif. Intell.*, vol. 3, no. 3, May 2021, Art. no. e200126.

[34] K. S. Gill and R. Gupta, "Chronic kidney disease detection using GridSearchCV cross validation method," in *Proc. Int. Conf. Recent Adv. Electr., Electron. Digit. Healthcare Technol. (REEDCON)*, May 2023, pp. 318–322.

[35] H. A. Mohamad Tabish, S. M. Arbaz, M. Agarwal, and A. Sinha, "Early prediction and risk identification of chronic kidney disease using deep learning technique," in *Proc. 15th Int. Conf. Comput. Commun. Netw. Technol. (ICCCNT)*, Jun. 2024, pp. 1–9.

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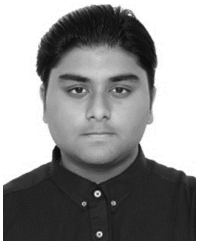

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